

Reconstructing ancestral gene regulatory toolkits through comparative genomics of unicellular Holozoa

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Diego Montiel Ayora

Omics Data Analyses

Master in Bioinformatics

Name of the tutor:

Marta Álvarez-Presas

Name of the SRP:

Xavier Grau-Bové

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Universitat Oberta
de Catalunya



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Name of the author:	Diego Montiel Ayora
Name of the tutor:	Marta Álvarez-Presas
Name of the SRP:	Xavier Grau-Bové
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Abstract

Understanding the origin of complex multicellularity in animals remains a central challenge in evolutionary biology. Developmental transcription factors (DTFs) play essential roles in cell differentiation and body-plan patterning in Metazoa. Although many DTFs were once considered animal-specific, recent studies on unicellular Holozoa, the closest living relatives of animals, have revealed that several homeobox families emerged earlier than previously assumed. In this study, we investigate the evolutionary history and expression patterns of non-TALE homeobox genes across representative unicellular holozoan lineages. We developed a customised bioinformatic pipeline integrating HMMER and BLAST searches to detect candidate homeodomain-containing sequences. The evolutionary history of these candidates were validated using Maximum Likelihood phylogenies. Our results indicate that non-TALE homeobox genes are more diverse and widely distributed among unicellular Holozoa than previously recognised, suggesting an unexpectedly deep evolutionary origin for key developmental regulators. These findings provide new insights into the genomic toolkit that predate animal multicellularity and underscore the early emergence of transcriptional complexity in holozoan evolution.

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Chapter 1

Introduction

1.1 Context and justification of work

The transitions from unicellularity to multicellularity are some of the most consequential innovations in evolution, allowing for new levels of biological complexity such as cell-cell communication and developmental regulation (Maynard Smith & Szathmáry, 1995). Multicellularity has evolved repeatedly across eukaryotes, and a particularly interesting case is the emergence of animals (Metazoa) as it gave rise to highly integrated multicellular organisms with complex developmental programs and extensive cell-type diversification (Ruiz-Trillo et al., 2023). When studying these events, one key piece is understanding the gene-regulation mechanisms that enable the coordination of cellular differentiation and patterning across tissues. In this project, we will focus on the origin and early evolution of key regulatory gene families in the lineage leading to animals.

1.1.1 The Origin of Multicellularity in Metazoa

Multicellular organisms can be divided by several categories. When considering biological complexity, multicellular organisms are often distinguished into simple and complex forms. Simple multicellularity refers to the organisms that arise from the mitotic divisions of a single progenitor, filaments, clusters, etc (Andrew H. & Knoll, 2011). They all have in common the presence of adhesive molecules that connect the products of successive cell divisions to form cohesive structures and limited morphological organization. On the other hand, in simple multicellular organisms, cell-to-cell communication and transfer of resources is either limited, non-specialized, or not developmentally regulated. Complex multicellular organisms show not only evidence of cell-cell adhesion but also intercellular communication and, commonly but with some exceptions, tissue differentiation mediated by networks of regulatory genes (Andrew H. & Knoll, 2011). When taken all this into consideration, animals are a perfect example of complex multicellular organisms, where the evolution of developmentally controlled processes made it possible to develop different types of tissues as well as body plans.

The origin of animal multicellularity is not only due to the appearance of new cell types, but by innovations in the regulation of gene expressions. The expansion of transcriptional factor families, the diversification of regulatory elements, and increase in complexity of developmental gene regulatory networks were involved in the emergence of Metazoa. All these innovations allowed for complex multicellularity to arise. Importantly, many of the components that regulate development in animals have been originated before the origin of Metazoa (Paps J, 2018; de Mendoza A. & Seb -Pedr s A., 2019).

Unicellular Holozoa

Unicellular holozoans are the closest extant protistan relatives of animals and, together with Metazoa, form the clade Holozoa. These organisms comprise several phylogenetically distinct lineages, including choanoflagellates, filastereans, ichthyosporeans and corallochytreans, and display diverse morphologies and life cycles that capture key features relevant to the origin of animal multicellularity. Choanoflagellates possess a single apical flagellum surrounded by a microvillar collar and can form clonal multicellular rosettes or alter colonial morphology in response to environmental cues. Filastereans, such as *Capsaspora owczarzaki*, exhibit amoeboid stages with filopodia-rich morphologies, transient aggregation, and secretion of extracellular matrix components, whereas ichthyosporeans often develop as multinucleate coenocytes that later cellularize to release daughter cells (de Mendoza & Seb -Pedr s, 2019; Ros-Rocher et al., 2021).

Comparative genomic and functional studies have revealed that unicellular holozoans possess many components of the molecular toolkit previously thought to be exclusive to animals. These include transcription factors and regulatory systems such as T-box/Brachyury (with conserved function in *Capsaspora owczarzaki*), RUNX, NF- B, Myc-Max networks, forkhead and homeobox gene families, as well as adhesion and signaling modules including cadherins, integrins, Notch/Delta and receptor tyrosine kinases (Ros-Rocher et al., 2021; Ruiz-Trillo & de Mendoza, 2020). Together, these findings suggest that the last unicellular ancestor of animals already harbored a gene-rich regulatory repertoire. At the same time, canonical animal distal enhancer architectures appear reduced or absent in at least some unicellular holozoans, indicating that key aspects of animal gene regulation were elaborated later along the metazoan stem (Ros-Rocher et al., 2021).

The combination of complex life cycles, transient multicellular behaviors, and the presence of extensive regulatory toolkits positions unicellular holozoans as a central comparative framework for investigating the origin of animal development. These organisms provide direct insight into how ancient regulatory modules were repurposed and integrated during the transition from unicellular ancestors to stable, developmentally regulated multicellularity in animals (Ros-Rocher et al., 2021; Ruiz-Trillo & de Mendoza, 2020).

1.1.2 The Homeobox gene family

Developmental transcription factors (DTFs) are DNA-binding proteins, sequence-specific, and play a role in the transcription of other proteins providing the establishment and execution of embryonic gene regulation networks. They are able to control the spatial and temporal expression of downstream target genes and, therefore, regulate developmental processes like cell-fate specification, body-axis patterning, and organogenesis. DTFs contain typically conserved DNA-binding domains such as the homeodomain, forkhead domain, or zinc-finger motifs. Evolutionary studies indicate that many TF families originated before the emergence of animals and subsequently diversified, providing the regulatory flexibility required for the evolution of complex multicellular development in Metazoa (Cheatle Jarvela A. M. & Hinman V. F., 2015; de Mendoza, A., & Seb  -Pedr  s, A., 2019).

Homeobox genes in particular are one of the most prominent families of DTFs in animals. Their proteins contain a conserved homeodomain of approximately 60 amino acids, that forms a three-helix helix–turn–helix fold. This third helix is responsible for the sequence-specific DNA recognition, although an N-terminal basic arm normally contributes forming additional contacts within the DNA. The homeodomain provides a shared DNA-binding framework but functional specificity among different homeobox proteins is achieved through class-specific motifs, dimerization, interactions with cofactors, and integration into larger regulatory complexes (Holland, P. W. et al, 2007).

In animals, homeobox genes are classified into two major superclasses: non-TALE (formed by ANTP, PRD, LIM, POU, CUT, HNF, SIX/SO, ZF, and CERS) and TALE superclass. The human genome contains approximately 235 functional homeobox genes distributed among these classes. Collectively, homeobox genes act as central regulators of developmental gene regulatory networks, controlling processes such as axial patterning, tissue and organ specification, neuronal differentiation, and chromatin-level regulation. Their deep evolutionary conservation and repeated involvement in developmental innovation underscore their importance in the emergence and diversification of animal body plans (Holland, P. W. et al, 2007; B  rglin, T. R., 2011; Holland, P. W., 2013).

Non-TALE homeobox genes form a diverse group of DTFs that share a homeodomain but lack a characteristic three–amino-acid insertion that are identity of the TALE superclass (Holland, P. W. et al, 2007). Comparative analyses indicate that most non-TALE classes originated early in animal evolution and subsequently diversified through gene duplication and domain rearrangement (Appendix A; Holland, P. W., 2013). For this reason they are an important piece in the origin of animal multicellularity puzzle.

The ANTP class: HOX and NKL subclasses

The ANTP class represents one of the largest groups of non-TALE homeobox genes and includes two major subclasses of particular developmental relevance: HOX and NKL (NK-like) genes. Members of the ANTP class are defined by a canonical homeodomain (Figure 1.1) and often possess short conserved motifs that mediate interactions with cofactors, contributing to com-

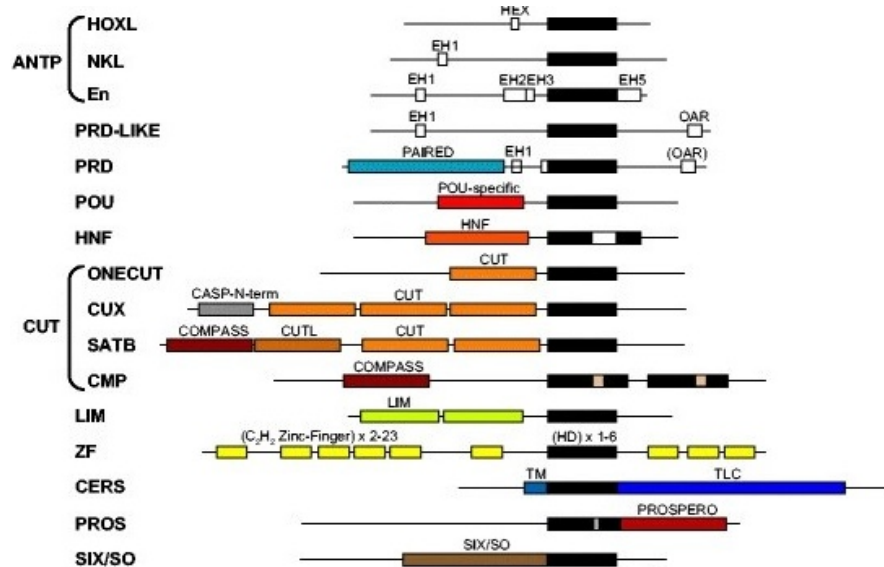


Figure 1.1: Non-Tale Homeobox genes and their respective domains. Extracted and edited from Bürglin, T. R., & Affolter, M. (2016).

binatorial transcriptional regulation. Phylogenetic reconstructions suggest that ANTP genes arose early in metazoan evolution and diversified prior to the emergence of complex animal body plans (Holland, P. W. et al, 2007; Holland, P. W., 2013).

HOX genes form a distinct subclass within ANTP and are characterized by their organization into genomic clusters. HOX proteins typically contain a conserved hexapeptide motif that facilitates interaction with TALE-class cofactors, enhancing DNA-binding specificity. Functionally, HOX genes are central regulators of anterior–posterior patterning and segmental identity, and their expansion through cluster duplication has been closely associated with increasing morphological and regulatory complexity in vertebrates (Holland, P. W. et al, 2007; Holland, P. W., 2013). HOX genes are present in Cnidaria, Placozoa and Bilateria (Ryan, J. F. et al, 2010).

The NKL subclass comprises several gene families, including NKX, DLX, MSX, TLX and related groups. NKL homeobox proteins share subclass-specific features within the homeodomain and frequently act as transcriptional repressors. Unlike HOX genes, NKL genes are generally not clustered, reflecting differences in genomic organization and regulatory deployment. Evolutionary analyses indicate that NKL genes originated early and diversified independently of HOX clusters and are present all across Metazoa, from Porifera and Ctenophora to Bilateria (Larroux, C. et al, 2007). Functionally, NKL genes are involved in a wide range of developmental processes, including neural patterning, organogenesis and cell-type specification, highlighting their broad contribution to developmental gene regulatory networks beyond axial patterning (Holland, P. W. et al, 2007; Holland, P. W., 2013).

LIM homeobox genes

LIM homeobox genes encode transcription factors that combine a C-terminal homeodomain with one or two N-terminal LIM zinc-binding domains (Figure 1.1). These LIM domains mediate protein–protein interactions, allowing LIM homeobox proteins to function as scaffolds within multiprotein regulatory complexes rather than acting solely through DNA binding. LIM homeobox genes are conserved across metazoans and their functions are closely associated with neuronal differentiation, tissue patterning and the specification of distinct cell identities (Holland, P. W. et al, 2007; Holland, P. W., 2013). LIM can be found all across Metazoa and also in Filasterea, Ichthyosporea, and Corallochytraea. No LIM has been found in Choanoflagellates (Grau-Bove, X. et al, 2017; Crespo Selma, B., 2024).

POU homeobox genes

The POU class is characterized by a bipartite DNA-binding region composed of a POU-specific domain linked to a POU-type homeodomain (Figure 1.1). This configuration enables flexible recognition of DNA motifs and supports combinatorial interactions with other transcription factors. Functionally, POU genes are involved in key developmental decisions, including cell-fate determination and the maintenance of cellular identity, reflecting their central position within developmental regulatory networks (Holland, P. W. et al, 2007; Holland, P. W., 2013).

CUT, HNF, SIX/SINE, ZF and CERS classes

The CUT class comprises homeobox genes encoding one or more CUT domains in combination with a homeodomain (Figure 1.1). These additional DNA-binding modules contribute to regulatory specificity and functional diversity. CUT homeobox genes are involved in transcriptional regulation during development (Holland, P. W. et al, 2007; Holland, P. W., 2013).

HNF homeobox genes represent a small and specialized class characterized by atypical homeodomain-like regions (Figure 1.1) and distinct dimerization properties. Their functions are primarily associated with endoderm-derived tissues and metabolic regulation (Holland, P. W. et al, 2007; Holland, P. W., 2013).

The SIX/SINE oculis class is defined by the presence of a conserved SIX domain upstream of the homeodomain (Figure 1.1), mediating interactions with cofactors. SIX genes are widely conserved among animals and are involved in sensory organ development and tissue specification (Holland, P. W. et al, 2007; Holland, P. W., 2013).

Finally, the ZF and CERS classes comprise smaller groups of non-TALE homeobox genes with distinctive domain architectures, including zinc-finger motifs or class-specific conserved regions (Figure 1.1). They contribute to regulation within animal developmental systems (Holland, P. W. et al, 2007; Holland, P. W., 2013).

The SOX gene family

The SOX (SRY-related HMG-box) gene family contains a series of developmental transcription factors defined by a conserved high mobility group (HMG) DNA-binding domain. SOX proteins play key roles in developmental processes such as cell-fate specification and maintenance of cellular identity and frequently act in combination with other transcription factors, particularly members of the POU family. Although SOX genes are central components of developmental gene regulatory networks, they do not belong to the homeobox superfamily, as they lack a homeodomain (de Mendoza et al., 2019).

Comparative genomic analyses have shown that SOX TFs originated prior to the emergence of Metazoa and are present in unicellular holozoans, indicating that elements of developmental transcriptional regulation predate animal multicellularity (Gao, Y. et al, 2024). For this reason, SOX genes were included in this study as a complementary reference gene family, providing a useful benchmark for evaluating the performance of the screening pipeline and for comparing the evolutionary behavior of non-homeobox developmental TFs with that of non-TALE homeobox genes.

1.2 Work objectives

1.2.1 General objectives

1. Identify developmental transcriptional factors (DTFs) in unicellular Holozoa.
2. Reconstruct evolutionary history of these DTFs for inferring how the genomic toolkit of the ancestral animal was.
3. Develop and test a bioinformatic pipeline that allows to reproduce all these steps (phylogenetic analyses, database construction...)

1.2.2 Specific objectives

1. Compile a comprehensive dataset including reference sequences of the DTF families of interest (e.g., LIM) together with the complete proteomes of selected target species and appropriate control species.
2. Identify candidate DTF sequences by performing domain and motif-based searches using a custom bioinformatic pipeline across the compiled proteomes.
3. Reconstruct phylogenetic relationships between reference and candidate sequences in order to evaluate their evolutionary history and classify them within established DTF families.
4. Validate the retrieved sequences by analyzing their phylogenetic placement and domain architecture.

1.3 Impact on sustainability, ethics-social and diversity

This project contributes to sustainability and ethical research goals by advancing fundamental knowledge on the origin and evolution of animal multicellularity. By improving our understanding of biodiversity, evolutionary innovation, and the emergence of biological complexity, this work aligns with Sustainable Development Goal (SDG) 15: Life on Land and SDG 14: Life Below Water, as many unicellular holozoans inhabit marine environments and constitute an important yet understudied component of oceanic biodiversity.

In addition, the development and application of reproducible bioinformatic pipelines and computational workflows directly supports SDG 9: Industry, Innovation and Infrastructure, by promoting open, scalable, and resource-efficient research practices. Computational analyses reduce the need for resource-intensive laboratory experimentation and facilitate data reuse, reproducibility, and methodological transparency.

Finally, this project contributes to ethical and inclusive scientific practice by focusing on underrepresented unicellular lineages that are often overlooked in studies of animal evolution. By expanding the taxonomic and conceptual scope of evolutionary research, the study fosters a more comprehensive and equitable understanding of biological diversity.

1.4 Focus and method followed

The main objective of this work was to identify and characterize developmental transcription factors (DTFs) in unicellular holozoans in order to gain insight into the evolutionary origin of the animal regulatory toolkit. Several methodological strategies could, in principle, be applied to address this objective. A first possibility would consist of performing simple similarity-based searches, such as BLAST-only approaches, to identify candidate homologs in target proteomes. Although straightforward and simple, this is often insufficient when working across deep evolutionary distances, as sequence divergence can obscure homology relationships. Alternatively, laboratory-based *in vitro* or *in vivo* functional experiments could be employed to assess the role of specific genes in unicellular organisms; however, these approaches are experimentally demanding, require suitable model systems, and are not feasible for broad comparative surveys.

In this study, a computational strategy was chosen that combines the adaptation of an existing domain-based screening pipeline with the development of additional downstream analyses. Specifically, an established pipeline was used to identify candidate sequences based on conserved protein domains, while custom scripts were developed to further filter results, analyze phylogenetic trees, and construct the final datasets.

The focus on proteome-wide analyses was motivated by the central role of conserved domains in defining DTF families. By targeting characteristic domains rather than relying exclusively on global sequence similarity, this strategy increases sensitivity to highly divergent homologs that may no longer be detectable through standard similarity searches alone. Moreover, the integration of phylogenetic reconstruction as a validation step enables candidate sequences to be evaluated in an explicit evolutionary framework, reducing the risk of false positives and allowing ambiguous cases to be identified and discussed.

Overall, the chosen strategy represents a middle-point between sensitivity and robustness for the study of DTFs in unicellular holozoans. It is particularly well suited to uncovering deeply conserved regulatory genes and assessing their evolutionary relationships, thus directly addressing the objectives of reconstructing the early history of developmental transcriptional regulation in the lineage leading to animals.

1.5 Work schedule

The development of this project followed a structured work plan, organized into sequential tasks distributed over a defined timeline. The planning was designed to allow iterative refinement of analyses, integration of results, and progressive validation of candidate developmental transcription factors.

1.5.1 Planned tasks

The main tasks carried out during the project included:

- Identification of developmental transcription factors (DTFs) across selected proteomes using a domain-based bioinformatic pipeline.
- Construction and analysis of phylogenetic trees to evaluate the evolutionary relationships of candidate DTF sequences.
- Integration of phylogenetic and presence–absence results to generate a comprehensive map of the emergence, diversification and loss of the studied DTF families.

1.5.2 Project calendar

The work was distributed over the following timeline:

- **October:** Initial analysis of proteome datasets provided by the MCG laboratory, including the first rounds of DTF screening and preliminary phylogenetic reconstructions.
- **November:** Completion of detailed analyses for complex gene families, particularly ZF, HOX and NKL, including additional filtering steps and phylogenetic validation.
- **December:** Integration of all results into a comprehensive evolutionary framework, construction of summary datasets and figures, and writing of the Master's thesis.

1.5.3 Milestones

Several key milestones had to be achieved throughout the project:

- Generation of a first validated list of candidate DTF sequences retrieved from MCG and EukProt datasets.
- Construction of phylogenetic trees for all major DTF families and completion of preliminary validation of candidate sequences.
- Completion of an integrated summary of presence–absence patterns and inferred divergence events across the sampled species.
- Implementation of automated phylogenetic tree parsing and sequence filtering using custom R scripts to generate curated gene-family datasets from raw phylogenetic outputs.

1.6 Brief summary of products obtained

This project resulted in a set of outputs and computational resources that collectively support the study of the evolutionary history of developmental transcription factors in unicellular holozoans and early-branching metazoans and can be applied to any dataset. The main products obtained are summarized below:

- Phylogenetic trees reconstructed for the ten targeted gene families (LIM, CERS, CUT, HNF, POU, SINE, ZF, HOX, NKL and SOX), enabling evaluation of sequence homology, evolutionary relationships and family-specific patterns across the sampled taxa.
- A curated and integrated dataset containing all retrieved sequences, including their corresponding taxonomic assignments and gene family classifications, serving as the basis for downstream comparative analyses.
- A custom R script developed to automate phylogenetic tree parsing, sequence filtering, and assembly of the final datasets, ensuring reproducibility and facilitating future extensions of the analysis.
- A presence–absence matrix summarizing the distribution of all targeted developmental transcription factor families across major taxonomic groups, providing a compact overview of evolutionary conservation and lineage-specific patterns.
- A schematic evolutionary tree integrating phylogenetic and presence–absence information, illustrating the inferred emergence, retention, and potential loss of the targeted gene families across holozoan and metazoan lineages.

1.7 Brief description of other memory chapters

This thesis is structured into a series of chapters that present the methods used, the results obtained, and discuss their biological and evolutionary implications.

Chapter 2 (Materials and Methods): describes the datasets, tools, and analytical strategies used throughout the study. Section 2.1 details the taxon sampling strategy and the selection of query sequences, defining the biological scope of the analyses. Section 2.2 introduces the EvoSearch pipeline, outlining its design and functionality for domain-based screening and phylogenetic reconstruction. Section 2.3 explains the procedures used for phylogenetic tree analysis and the construction of the final curated dataset, including additional filtering steps required for complex gene families.

Chapter 3 (Results): presents the outcomes of applying the described pipeline and analytical framework. Section 3.1 focuses on the phylogenetic trees generated for the targeted gene families, highlighting differences between domain-specific and non-domain-specific cases. Section 3.2 examines the structure and composition of the resulting datasets, including the

identification and handling of repeated sequences (Section 3.2.1) and the analysis of the final curated dataset (Section 3.2.2), culminating in the presence–absence patterns across taxa.

Chapter 4 (Discussion): contextualizes the results within a broader evolutionary and methodological framework. Section 4.1 evaluates the performance and limitations of the domain-based screening pipeline, while Section 4.2 addresses the specific challenges encountered in reconstructing the evolutionary history of the ANTP class. Section 4.3 synthesizes the phylogenetic and distributional results to discuss the evolutionary distribution of non-TALE homeobox genes and their implications for the origin of animal regulatory complexity.

Chapter 5 (Conclusions and future works): summarizes the main findings of the thesis, reflects on the degree to which the proposed objectives were achieved, and outlines potential directions for future research building upon the results and resources generated in this study.

Chapter 2

Materials and methods

2.1 Taxon Sampling & Query selection

To conduct the screening of the different DTFs, we first defined the taxonomic sampling of species to be analyzed. The dataset includes all available unicellular holozoans curated at the MCG lab (<https://multicellgenome.com/genomes>), together with several recently described choanoflagellate species retrieved from the EukProt database (Richter, D. J. et al, 2022; <https://evocellbio.com/eukprot/>). In addition, a broad selection of early-branching metazoans, including ctenophores and sponges, was incorporated to capture key stages in early animal evolution. A subset of bilaterian species was also included as a control group, given their comparatively well-annotated genomes. Several fungal species were added as an outgroup for subsequent phylogenetic analyses (Figure 2.1). Finally, for each selected species, the complete predicted proteome was used as input for downstream analyses.

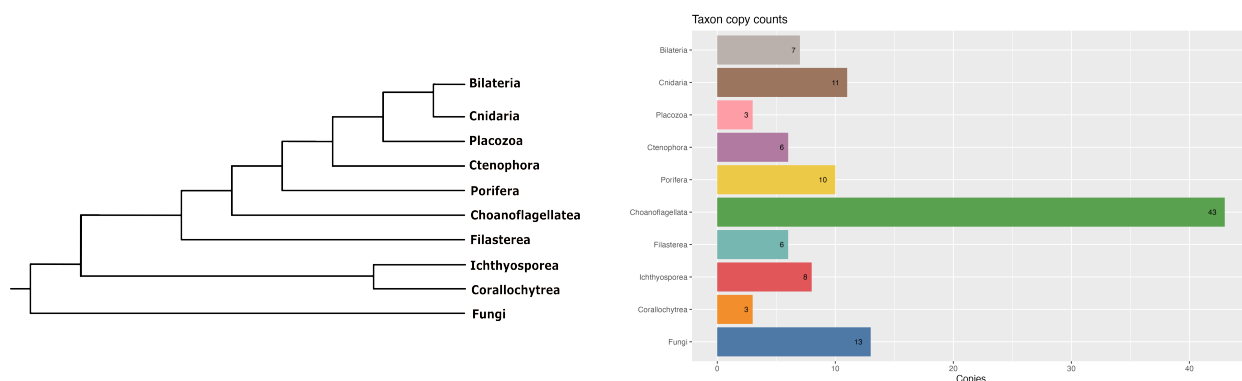


Figure 2.1: Left: tree representing the phylogenetic relationships across all Holozoa, with Fungi as an outgroup. Right: number of species per taxon group. The list of the specific species used is found in Appendix B

In addition to the proteome datasets, a set of query sequences was required for the screening analyses. Query proteins were retrieved from the NCBI (NCBI Resource Coordinators, 2023)

and UniProt (The UniProt Consortium, 2023) databases and consisted of representative members of the different subfamilies within each target gene family mostly from *Homo sapiens* (e.g. for the LIM class, ISL, LHX1–LHX9 and LMX1A were selected). For each gene family, the selected sequences were compiled into a FASTA file that was subsequently used as input for downstream analyses (Appendix C).

Finally, both the complete proteome datasets of the selected species and the query sequence files were subjected to header normalization. We replaced non-standard characters (e.g. "*", ".", "-") and whitespace in sequence headers with underscores ("_"). This step was required to prevent parsing errors in downstream pipeline steps and to ensure that full sequence annotations were correctly retained and displayed in the resulting phylogenetic trees.

2.2 EvoSearch pipeline

The pipeline is used to screen the target proteomes, named EvoSearch, has been developed at the Multicell Genome Lab (MCG, Institute of Evolutionary Biology) and is pending publication. It takes the input query sequence and performs a search of their domains, as well as a search by similarity, on the targeted proteomes. Its main characteristic is that it is domain-centric and this allows the use of permissive similarity thresholds during sequence retrieval, enabling the recovery of divergent homologs while ensuring that the defining domains of the target gene families are retained.

Snakemake was used to create the pipeline and, prior to running it, several parameters must be specified, including the Pfam identifiers (Mistry et al., 2021) of the target domains, the e-value threshold to be applied by HMMER (Katoh & Standley, 2013) and BLAST (Altschul et al., 1997), and the query and proteome datasets to be analyzed.

The pipeline consists of the following main steps:

1. Alignment of the query sequences using MAFFT (Katoh & Standley, 2013).
2. A filtering step retains only sequences containing the targetted Pfam domains, as detected using HMMER.
3. The target proteomes fetched sequences are aligned using MAFFT.
4. A second filtering step performs similarity searches using BLAST and HMMER, applying the specified e-value threshold to refine the set of candidate sequences.
5. A phylogenetic tree is then inferred from a MSA matrix obtained from the final set of retrieved sequences using IQ-TREE (Nguyen et al., 2015).

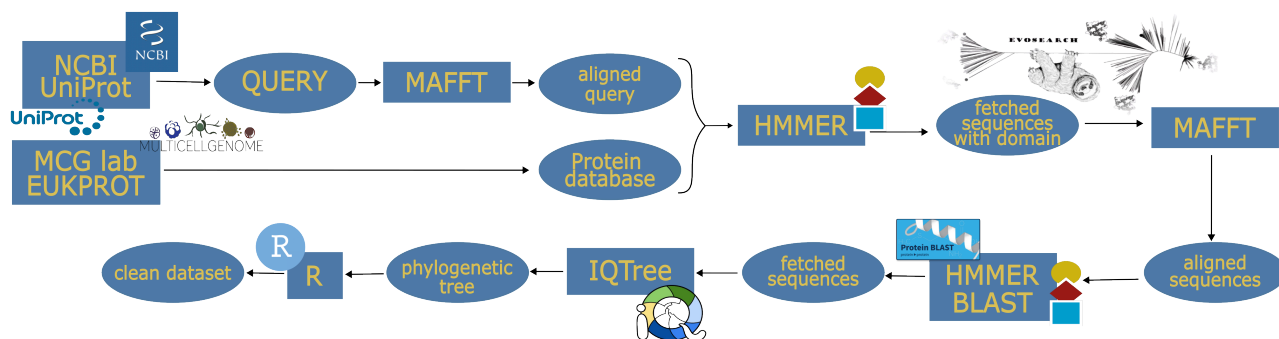


Figure 2.2: Diagram summarizing the overall pipeline. The figure includes the preparation of query and proteome datasets, the domain-based screening and filtering steps, and the generation of phylogenetic trees. In some cases, downstream phylogenetic analyses were subsequently performed in R (Section 2.3).

2.3 Phylogenetic tree analyses & creation of the final dataset

In cases where the target gene family is characterized by highly specific domains (e.g. LIM genes and LIM domains), the pipeline output can be directly interpreted from the resulting phylogenetic tree. Under these conditions, candidate sequences are sufficiently well resolved, and downstream analyses focus on extracting the species composition and corresponding sequences from the tree. To achieve this, a custom R script was used to:

1. Load the inferred phylogenetic tree file.
2. Extract the terminal nodes (tips) corresponding to individual protein sequences.
3. Match the extracted sequence headers against a reference annotation dataset in order to infer species identity.
4. Store the resulting information in a gene-specific dataset.

In contrast, for gene families lacking highly specific domains (e.g. SOX genes), the permissive similarity thresholds applied during the pipeline allow the retrieval of additional proteins sharing the same conserved domain but not belonging to the target gene family. In these cases, further phylogeny-based filtering was required. A second custom R script was developed to perform the following steps:

1. Load the inferred phylogenetic tree file.
2. Identify tree tips corresponding to sequences correctly annotated as members of the target gene family.
3. Retain these validated sequences together with all neighbouring tips located within n branches in the phylogeny. The value of n was chosen to ensure the inclusion of closely related sequences while minimizing the retention of distant paralogs and unrelated proteins.
4. Match the selected sequence headers to the reference annotation dataset and store the results in a gene-specific dataset.

This procedure was applied independently to all analyzed gene families using a single unified script (Appendix D). The resulting gene-specific datasets were subsequently merged into a final dataset, which was used for downstream comparative and evolutionary analyses.

Chapter 3

Results

We have retrieved 2 sets of results:

1. Phylogenetic trees for each target gene.
2. A full dataset with all the inferred information.

3.1 Phylogenetic trees

All phylogenetic trees generated in this study have been edited for visualization and interpretation purposes using the Interactive Tree Of Life platform (iTOL; Letunic & Bork, 2021). Trees were midpoint-rooted to provide a consistent and comparable representation across gene families. In order to improve readability, well-supported and clearly annotated clades were collapsed when appropriate, while maintaining the overall topology relevant to the evolutionary questions addressed. Branch support values were retained, and nodes with high support (bootstrap values ≥ 90) were highlighted. Additionally, sequences from unicellular holozoans were colored across all trees to facilitate their identification and to visually assess their phylogenetic placement relative to metazoan sequences.

LIM

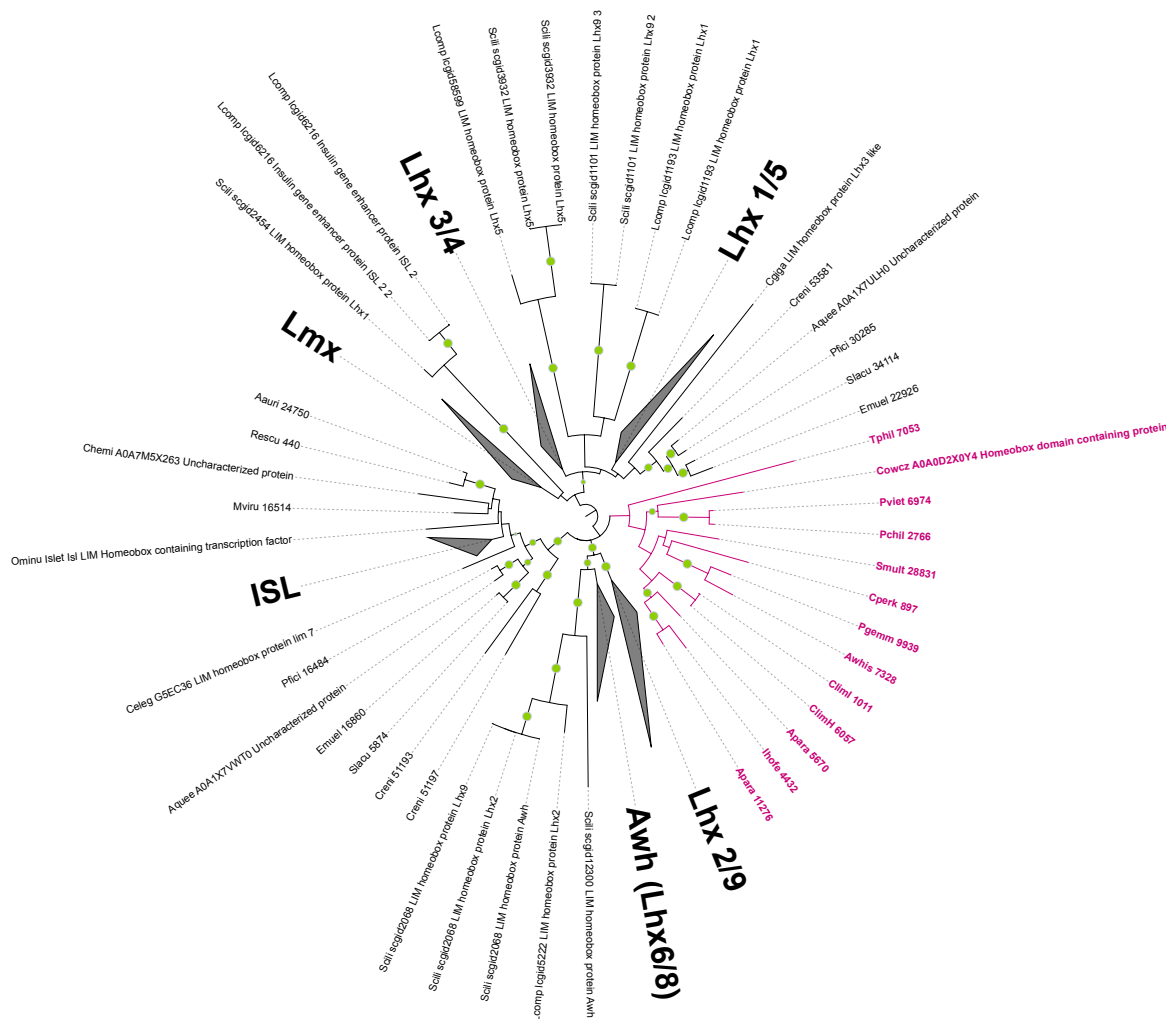


Figure 3.1: Phylogenetic tree retrieved from the pipeline using LIM as the targeted gene. In purple we have colored all unicellular Holozoa present. The green circles in the nodes indicate that the support is over 90. The tree was generated using Maximum-Likelihood approach in IQ-TREE. The best-fit amino-acid substitution model was selected using ModelFinder ((Q.insect+F+I+G4). Branch support values correspond to ultrafast bootstrap support (UF-Boot; 1000 replicates) and indicate support for each internal node with values ranging from 0 to 100 (Nguyen et al., 2015; Kalyaanamoorthy et al., 2017; Hoang et al., 2018).

The phylogenetic reconstruction of LIM homeobox genes resulted in a well-resolved tree in which sequences clustered according to the expected LIM subclasses (Figure 3.1). Metazoan sequences formed clearly defined clades, consistent with established family assignments. In addition, several unicellular holozoan sequences were recovered and grouped within the LIM homeobox clade, rather than appearing scattered across the tree.

CERS

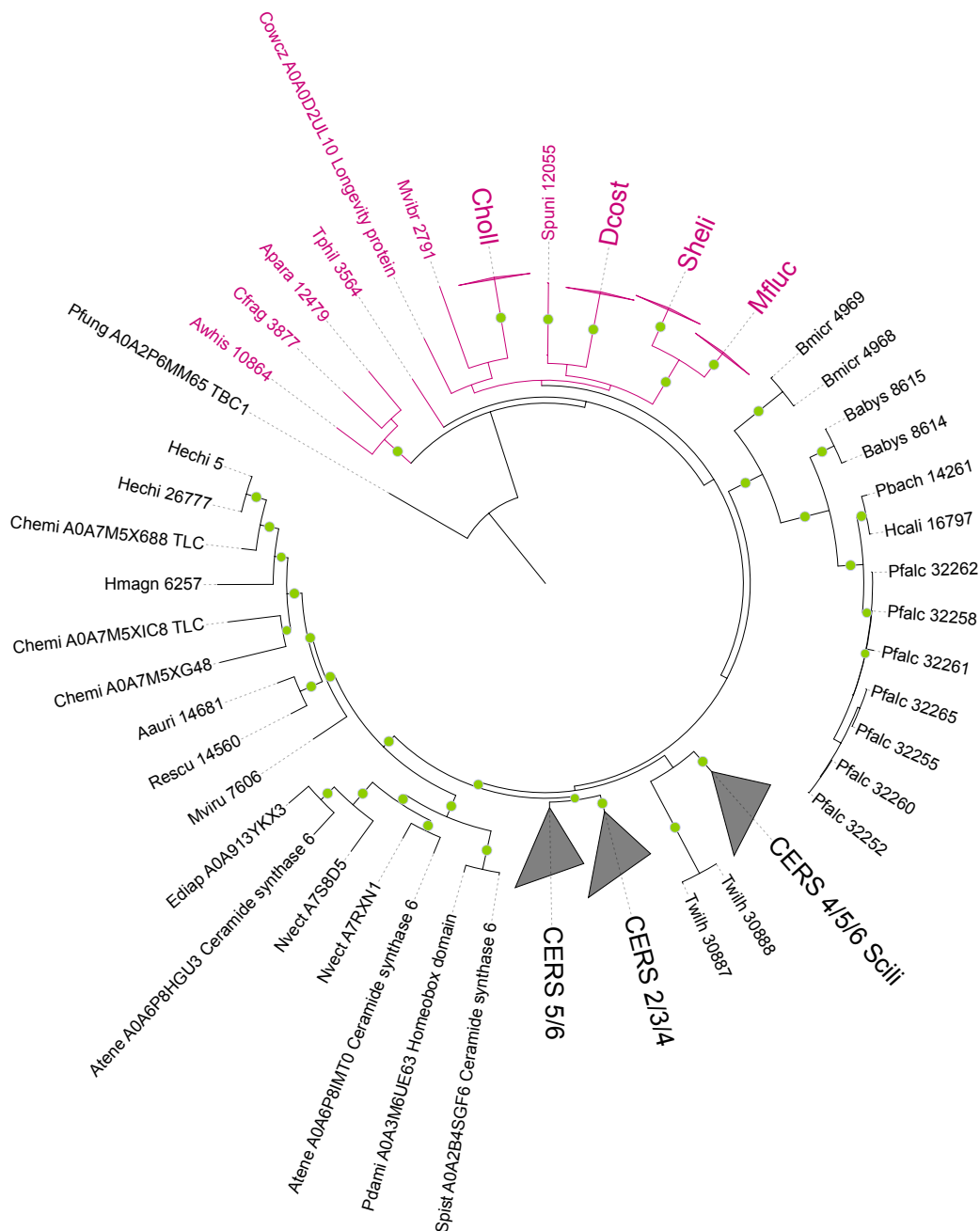


Figure 3.2: Phylogenetic tree retrieved from the pipeline using CERS as the targeted gene. In purple we have colored all unicellular Holozoa present. The green circles in the nodes indicate that the support is over 90. The tree was generated using Maximum-Likelihood approach in IQ-TREE. The best-fit amino-acid substitution model was selected using ModelFinder (Q.pfam+I+G4). Branch support values correspond to ultrafast bootstrap support (UFBoot; 1000 replicates) and indicate support for each internal node with values ranging from 0 to 100 (Nguyen et al., 2015; Kalyaanamoorthy et al., 2017; Hoang et al., 2018).

The phylogenetic reconstruction of CERS homeobox genes resulted in a compact and well-resolved tree (Figure 3.2). Retrieved sequences clustered tightly within the CERS family, and several unicellular holozoan sequences were recovered and grouped together.

POU

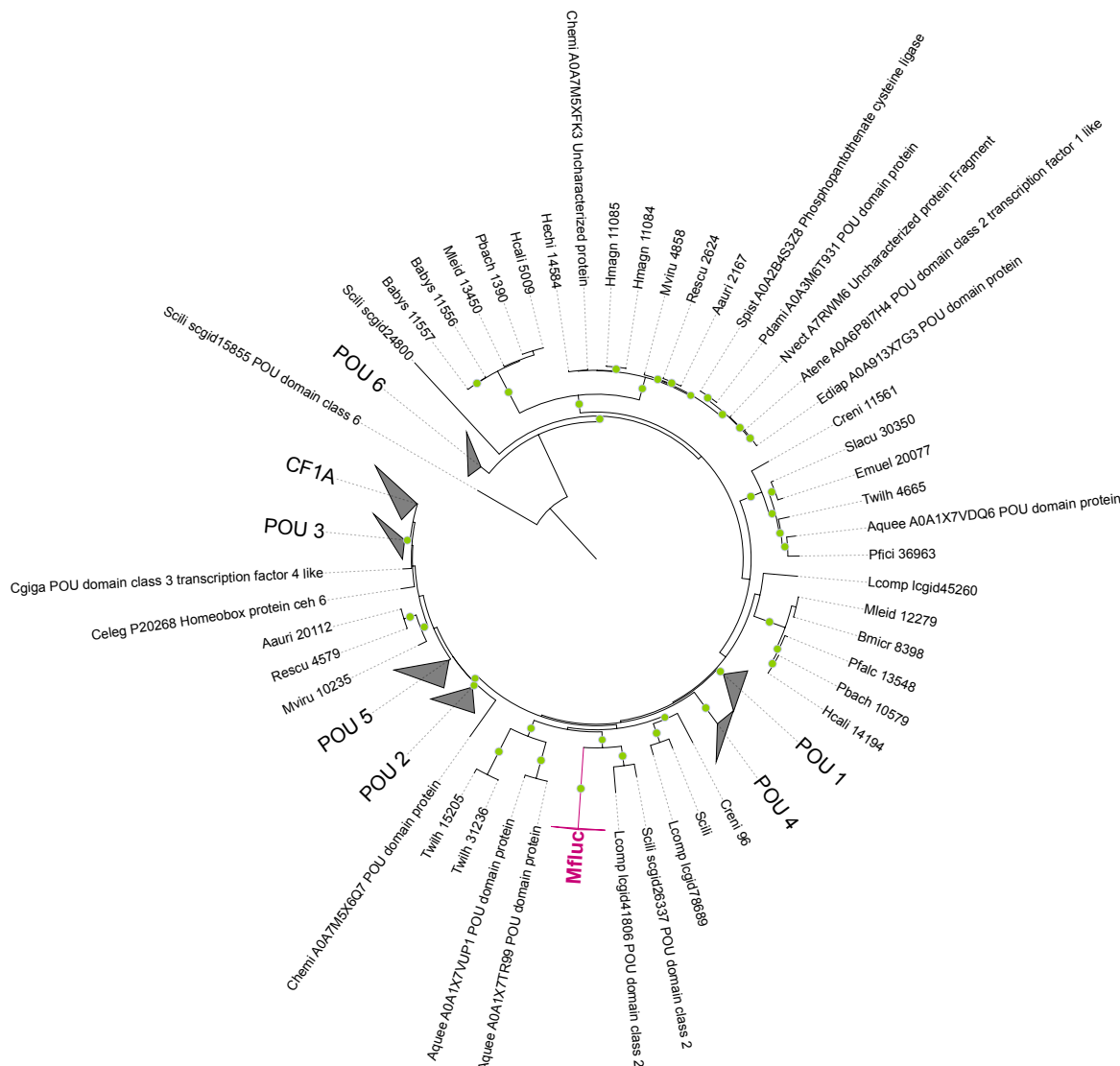


Figure 3.3: Phylogenetic tree retrieved from the pipeline using POU as the targeted gene. In purple we have colored all unicellular Holozoa present. The green circles in the nodes indicate that the support is over 90. The tree was generated using Maximum-Likelihood approach in IQ-TREE. The best-fit amino-acid substitution model was selected using ModelFinder (Q.insect+F+I+G4). Branch support values correspond to ultrafast bootstrap support (UF-Boot; 1000 replicates) and indicate support for each internal node with values ranging from 0 to 100 (Nguyen et al., 2015; Kalyaanamoorthy et al., 2017; Hoang et al., 2018).

The POU gene family produced a well-structured phylogenetic tree with clear separation of major POU subclasses (Figure 3.3). Bilaterian and early-branching metazoan sequences clustered as expected, forming coherent clades. Several unicellular holozoan sequences were recovered and grouped together.

CUT, HNF & SINE

Phylogenetic analyses of the CUT, HNF and SINE homeobox gene families produced well-resolved and internally consistent trees, with sequences clustering according to established family relationships (Appendix E). In all three cases, retrieved sequences were restricted to metazoan taxa and formed coherent clades corresponding to known subclasses within each family. No unicellular holozoan sequences were detected for these gene families, indicating that the domain-based screening approach did not retrieve spurious non-metazoan homologs and supporting a metazoan-restricted distribution for CUT, HNF and SINE homeobox genes in the sampled taxa.

ANTP, ZF and SOX

In contrast to the gene families characterized by highly specific domains, phylogenetic reconstruction of SOX, ZF and the ANTP superclass (both HOX and NKL) TFs produced a markedly complex and poorly resolved tree (Appendix E). In the case of SOX, its HMG domain is shared by multiple transcription factor families, resulting in the retrieval of numerous non-SOX sequences despite the application of similarity-based filtering. Consequently, the inferred phylogeny contained a heterogeneous assemblage of proteins, with extensive intermixing of target and non-target sequences. The rest of the phylogenetic trees (ZF and ANTP) also follow a similar profile although in these cases almost all different homeobox gene families were retrieved.

This lack of domain specificity and the resulting phylogenetic ambiguity necessitated the implementation of additional phylogeny-based filtering steps to reliably identify homologs, as described in the Materials and Methods section.

3.2 Dataset analyses

The final dataset contains 6 columns (Header, Code, Species, Gene, Category & Taxonomy), organizing each sequence (the rows) by its full header, the species it belongs to and different taxonomical categories and finally, the gene family it has being assigned.

3.2.1 Repeated sequences

First we noticed that when analyzing the full dataset, from 2298 rows, 369 of them had duplicated headers but the gene they had being assigned was different. This meant that the same sequence was being assigned the same gene. All of the cases had at least one the genes being flagged as does that didn't had gene family-specific domains (SOX, NKL, HOX and ZF), which made sense since their respective phylogetic trees had many sequences from other genes.

To address this issue we implemented two new filters:

1. If one of the sequences had already being flagged by a gene with a family-specific domain, it was assigned to that gene only and the rest of the rows deleted.
2. If the gene was already annotated, their annotated gene had prevalence and the rest of rows where also deleted.

After this last filtering we still had 181 duplicated sequences. Most of these remaining genes where pairs of the same sequence asociated both to HOX and NKL. This posed an interesting possibility though, by both being part of the ANTP super class, since it could mean that the conflicting results are due to these sequences being an ancestor to the whole ANTP super class. Even so, these duplicated headers where subtracted from the final dataset analyses.

3.2.2 Final dataset analyses

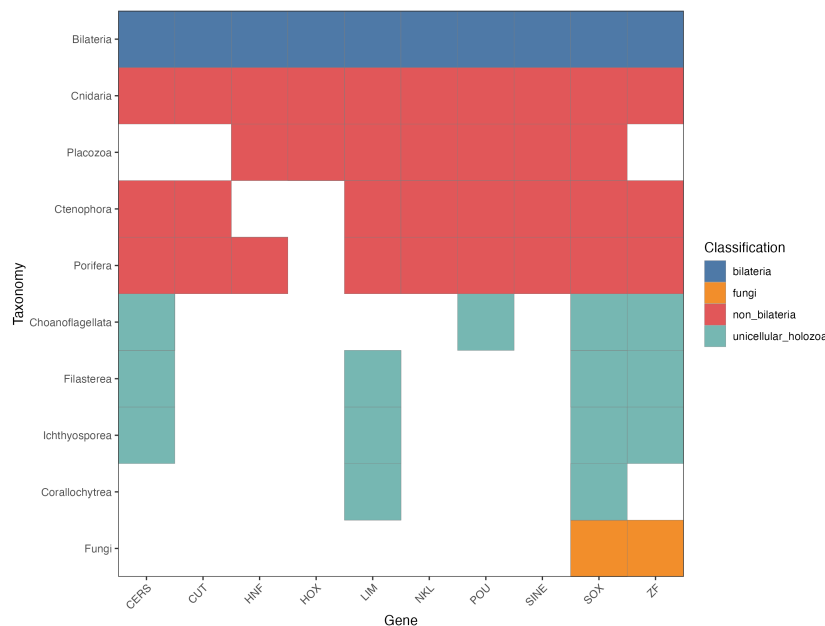


Figure 3.4: Presence (colored squares) or absence (white squares) matrix showing the distribution of each studied DTF family among unicellular holozoans and metazoan clades.

After all the tinkering we were able to build a presence-absence matrix (Figure 3.4) regarding the targeted genes and taxonomical groups. As expected, all analyzed gene families were detected across bilaterians and non-bilaterian metazoans. In contrast, the distribution of homeobox genes in unicellular holozoans was more heterogeneous. Several gene families (CUT, HNF, HOX, NKL and SINE) were not retained in unicellular holozoans, either due to absence or because additional analyses were required to resolve duplicated sequences, as in the case of HOX and NKL genes. LIM and POU homeobox genes were detected in a subset of unicellular holozoan lineages, indicating a restricted but detectable presence outside Metazoa. Notably, ZF homeobox genes were consistently identified across multiple unicellular holozoan taxa, representing the most broadly distributed non-TALE homeobox class outside animals in this dataset.

Chapter 4

Discussion

4.1 Performance and limitations of a domain-based screening pipeline

Evaluation of the pipeline performance revealed two distinct scenarios, depending on the domain architecture of the target gene families:

1. Gene families characterized by highly specific domains (e.g. LIM, CERS, SINE, POU, HNF and CUT).
2. Gene families defined by domains belonging to broader superclasses shared across multiple transcription factor families (e.g. HOX, NKL, ZF and SOX).

In the first scenario, the results were highly conclusive. Phylogenetic trees generated for gene families with specific domains displayed clear and coherent topologies, with sequences clustering according to known family assignments and no detectable inclusion of unrelated gene families. This indicates that the domain-centric strategy was sufficient to reliably identify homologous sequences across the sampled taxa.

In contrast, gene families belonging to the second scenario exhibited markedly different behavior. Phylogenetic trees inferred for these families consistently contained sequences from multiple gene families, and candidate target sequences were dispersed across the tree rather than forming well-defined clades. This pattern reflects the lack of domain specificity and highlights the limitations of similarity-based searches when applied to broadly conserved domains (Krishnamurthy, N. et al, 2007).

These results are consistent with the evolutionary context of the study. As outlined in the Introduction, the taxon sampling spans large evolutionary distances, including unicellular holozoans, early-branching metazoans and bilaterians. Analyses across such deep divergences are particularly susceptible to the retrieval of distantly related or functionally unrelated sequences when diagnostic domain features are absent.

Taken together, our results demonstrate that domain-centric screening pipelines are highly effective for gene families defined by specific and diagnostic domains, even across deep evolutionary timescales. Conversely, for gene families lacking such specificity, similarity-based approaches alone are insufficient and must be complemented by additional phylogeny-based filtering and manual curation to reduce artifacts and achieve reliable sequence classification.

4.2 Challenges in reconstructing ANTP-class evolution

The ANTP class represents a particularly challenging case in this study. Although HOX and NKL gene families were searched independently using family-specific query sequences, neither family is characterized by a uniquely diagnostic domain. Consequently, the screening relied on the presence of the homeobox domain alone, resulting in phylogenetic trees that included representatives of multiple homeobox gene families rather than exclusively HOX or NKL sequences.

Despite careful filtering of the dataset, as described in the Materials and Methods and Results sections, several sequences remained ambiguously assigned to both HOX and NKL families. Given that these two gene families belong to the same ANTP superclass, one possible explanation is that these sequences represent ancestral ANTP-like genes that predate the divergence of HOX and NKL lineages.

To explore this possibility, an additional phylogenetic analysis was performed using IQ-TREE (Nguyen et al., 2015). This analysis included: (i) annotated HOX and NKL sequences sampled across Metazoa, (ii) the ambiguous sequences retrieved from unicellular holozoans, (iii) annotated PRD sequences, which represent the sister group to the ANTP superclass, and (iv) LIM and CERS homeobox genes, included as outgroups to verify that the target sequences did not fall outside the broader homeobox gene clade.

In the resulting phylogeny (Figure 4.1), some of the ambiguous unicellular holozoan sequences were recovered within a clade next to PRD, a placement consistent with their potential assignment as ancestral ANTP-like genes or highly divergent PRD-related sequences. However, other ambiguous sequences clustered within LIM or CERS clades, despite lacking the characteristic domains of these families. In addition, several basal nodes displayed low support values, limiting confidence in the inferred deep relationships. This can be explained by big divergencies between families and due to some sequence fragments being too short and maybe not having enough phylogenetic signal.

Taken together, this analysis does not allow a definitive assignment of the ambiguous sequences to specific ANTP subclasses. While their phylogenetic positions are compatible with the presence of ancestral or highly divergent homeobox genes in unicellular holozoans, the limited resolution and low support values indicate that additional analyses will be required to resolve their evolutionary affinities. Such analyses may include expanded taxon sampling, domain architecture characterization, trying phylogenies based on domains (like ones done with AlphaFold) and complementary sequence-based approaches to determine whether these sequences represent bona fide homeobox genes or artifacts of deep phylogenetic divergence.

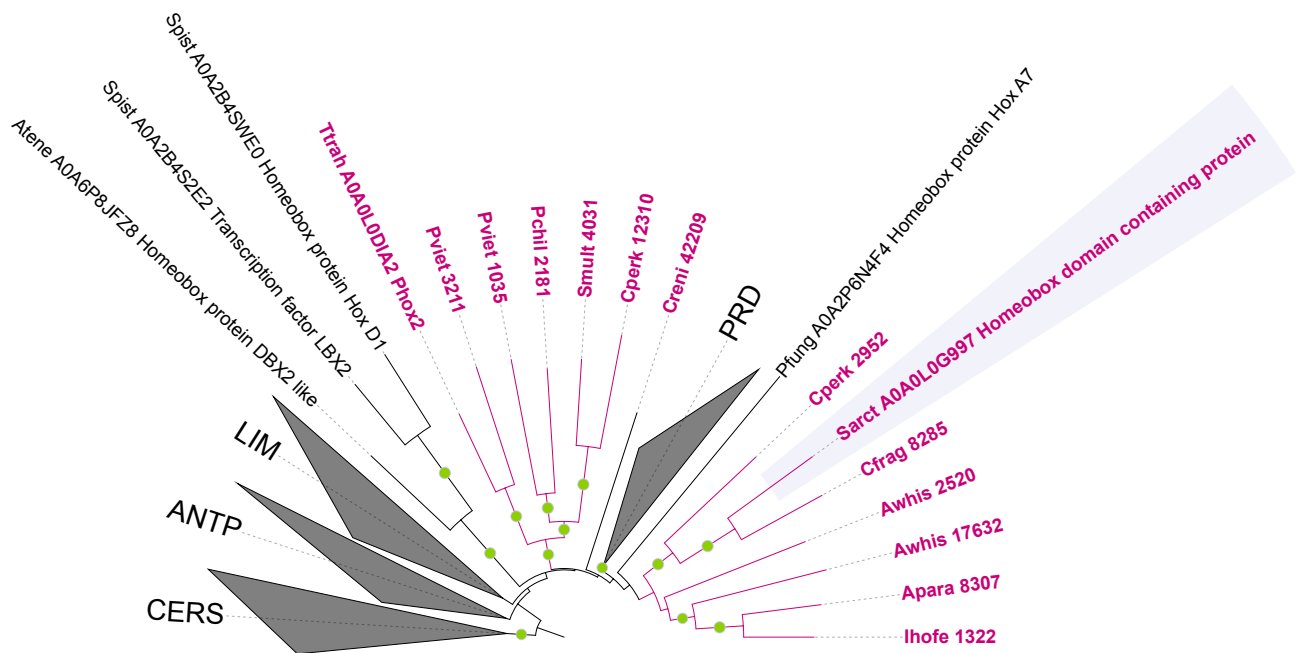


Figure 4.1: Phylogenetic tree inferred using IQ-TREE including annotated HOX, NKL, PRD, LIM and CERS sequences, together with ambiguous unicellular holozoan sequences. In purple we have colored all unicellular Holozoa present. The green circles in the nodes indicate that the support is over 90. The tree was generated using Maximum-Likelihood approach in IQ-TREE. The best-fit amino-acid substitution model was selected using ModelFinder (VT+F+I+G4). Branch support values correspond to ultrafast bootstrap support (UFBoot; 1000 replicates) and indicate support for each internal node with values ranging from 0 to 100 (Nguyen et al., 2015; Kalyaanamoorthy et al., 2017; Hoang et al., 2018). Edited in iTOL (Letunic & Bork, 2021)

4.3 Evolutionary distribution of non-TALE homeobox genes

To integrate the results obtained from phylogenetic reconstructions and presence–absence analyses, a schematic summary of the inferred evolutionary distribution of the analyzed non-TALE homeobox gene families was generated (Figure 4.2). This representation combines the curated dataset with the phylogenetic framework of major holozoan and metazoan lineages, highlighting putative points of gene emergence, retention, and loss.

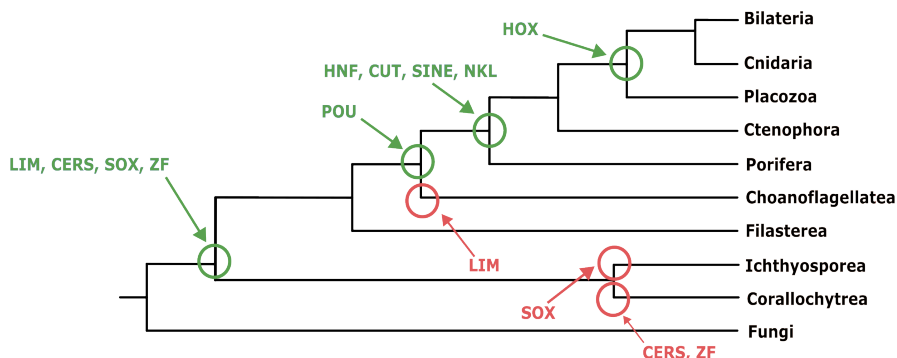


Figure 4.2: Phylogenetic reconstruction summarizing the inferred (green) origins and losses (red) of the studied DTF families across holozoan lineages.

Overall, the reconstructed distribution is largely consistent with current knowledge of homeobox gene evolution. Several gene families, including HNF, CUT, SINE and NKL, appear restricted to metazoans in the final curated dataset, with their presence emerging after the divergence of unicellular holozoans. The HOX gene family is inferred to have arisen later within Metazoa, consistent with its established role in axial patterning and its absence outside animals. These patterns support the view that a substantial expansion and specialization of homeobox genes occurred along the metazoan stem lineage.

In contrast, a subset of non-TALE homeobox gene families shows evidence of earlier origins. LIM and POU homeobox genes were detected in multiple unicellular holozoan lineages, suggesting that these families predate the emergence of animals and were already present in ancestral holozoans.

Notably, ZF homeobox genes represent the most striking exception to previously described distributions. In this study, ZF homeobox sequences were consistently identified across several unicellular holozoan lineages, indicating a broader evolutionary distribution than previously recognized. Although the functional roles of these genes in unicellular contexts remain unknown, their presence suggests that homeobox-mediated regulatory mechanisms may have deeper evolutionary roots than traditionally assumed.

One of the most noticeable patterns emerging from this study concerns the distribution of

LIM homeobox genes across holozoan lineages. LIM genes were detected all holozoans except from choanoflagellates, despite this group representing the closest known relatives of animals. This highlights the dynamic nature of gene-family evolution around the animal stem lineage. Previous comparative analyses indicate that gene-family turnover in this evolutionary window is high and that lineage-specific gene loss in close animal relatives is common. Consequently, the absence of LIM genes in choanoflagellates does not necessarily imply an animal-specific origin or methodological bias, but rather is consistent with secondary loss (Richter, D. J. et al, 2018; López-Escardó, D. et al, 2019).

From a functional perspective, LIM homeobox genes facilitate protein–protein interactions involved in cell identity specification, tissue patterning and differentiation. It is possible then that regulatory functions typically mediated by LIM homeobox genes in other holozoans are either not required in choanoflagellates or are carried out by alternative transcription factors or regulatory mechanisms. Choanoflagellates exhibit distinct life histories and modes of cellular organization compared to other unicellular holozoans, which may have relaxed selective constraints on the retention of LIM genes or favored functional replacement by other components of their regulatory networks. Under this scenario, the absence of LIM genes in choanoflagellates would reflect a lineage-specific regulatory change rather than a difference in the evolutionary origin of LIM homeobox genes (Fairclough, S. R., 2011).

Finally, the schematic representation also reflects instances where gene absence may result from lineage-specific loss or incomplete sampling rather than true evolutionary absence. As such, the inferred origins and losses depicted in Figure 4.2 should be interpreted as working hypotheses that summarize the current dataset rather than definitive evolutionary events. Together, these results highlight a diverse pattern of conservation and innovation in non-TALE homeobox gene families and show the importance of unicellular holozoans for reconstructing the early evolution of animal regulatory gene repertoires.

Chapter 5

Conclusions and future work

Conclusions

The main conclusions of this study can be summarized as follows:

1. The domain-based screening pipeline applied in this project proved to be a robust bioinformatic strategy for the detection of developmental transcription factor families characterized by highly specific and diagnostic domains. In these cases, the combination of domain annotation and Maximum Likelihood phylogenetic inference yielded well-resolved trees, enabling reliable identification of homologous sequences across deep evolutionary distances.
2. Analyses of gene families lacking unique family domains highlighted the limitations of similarity-based and domain-centric bioinformatic approaches when applied to highly divergent proteomes. These cases demonstrated an increased sensitivity to false hits and ambiguous classifications, underscoring the necessity of integrating phylogenetic validation, additional filtering criteria, and manual curation to ensure accurate sequence classification.
3. Comparative bioinformatic analyses of the final curated dataset revealed that several non-TALE homeobox developmental transcription factor families originated prior to the emergence of Metazoa. The detection of LIM, POU, and other non-TALE homeobox genes in unicellular holozoans supports the inference that substantial components of developmental gene regulatory networks were already present in premetazoan lineages.
4. Importantly, this work provides computational evidence for the presence of homeobox zinc-finger (ZF) genes in unicellular holozoans, extending the known evolutionary distribution of this gene family beyond Metazoa. This result illustrates the capacity of integrative bioinformatic pipelines combining domain detection, phylogenetic inference, and dataset curation to uncover previously unrecognized evolutionary patterns, and it highlights the potential of large-scale in silico screening to generate novel hypotheses for future functional validation.

Assessment of objectives

The objectives proposed at the beginning of this study were largely fulfilled, although with varying degrees of resolution depending on the gene family and analytical context. With respect to the general objectives, developmental transcription factors were successfully identified in unicellular holozoans, particularly for gene families characterized by specific domains, demonstrating that such regulatory components are not restricted to Metazoa. Phylogenetic and comparative analyses enabled the reconstruction of the evolutionary history of multiple non-TALE homeobox gene families, providing insight into the composition of the genomic toolkit present in the lineage leading to animals. While some gene families could not be assigned conclusively due to deep evolutionary divergence and limited phylogenetic resolution, these limitations were explicitly characterized and contextualized. In addition, a reproducible bioinformatic pipeline integrating sequence retrieval, domain-based screening, phylogenetic reconstruction was thoroughly tested and a series of scripts were developed for further phylogenetic analyses and dataset curation downstream.

Regarding the specific objectives, a comprehensive dataset was compiled, including complete proteomes of selected unicellular holozoans, early-branching metazoans, bilaterians and appropriate outgroups, together with curated reference sequences for each developmental transcription factor family. Candidate sequences were identified using domain and motif-based searches implemented in the custom pipeline, followed by phylogenetic reconstruction to evaluate their evolutionary relationships. The combination of phylogenetic placement and domain architecture analyses allowed the validation and classification of the majority of retrieved sequences. In cases where definitive classification was not possible, the sources of ambiguity were identified and discussed, highlighting both the strengths and limitations of the approach. Overall, this study met its stated objectives and provides a robust framework for future investigations into the evolutionary origins of animal developmental gene regulatory networks.

Assessment of the schedule and methodology

Overall, the work schedule defined at the beginning of the project was followed successfully, and the main planned tasks were completed within time. The identification of DTFs, phylogenetic reconstruction of the targeted gene families, and integration of the results into a comparative evolutionary framework were carried out as planned.

Our approach proved adequate for addressing the central aims of the study. In particular, the domain-based screening pipeline and subsequent phylogenetic analyses were effective for most gene families and provided robust results. However, the project was initially conceived to also include transcriptomic analyses of selected unicellular holozoans in order to assess gene expression patterns and try to infer possible functions of the unicellular holozoa homeobox genes. As the project has advanced, it has become clear that the complexity and workload associated with the genomic and phylogenetic components required a much more investment of time than initially calculated. For this reason, transcriptomic analyses were deemed unrealistic within the available time for the presentation of this project.

Finally, the scope of the project has expanded beyond the initial plan through the preparation of multiple poster and oral presentations and the establishment of a collaboration with another research group. While these activities increased the overall workload, they have also contributed to the scientific development of the project and to the divulgation and discussion of the results.

Assesment of the impact on sustainability, ethics-social and diversity

Overall, this project demonstrates that fundamental research in evolutionary genomics can contribute meaningfully to several Sustainable Development Goals. By improving knowledge of biodiversity and evolutionary processes in marine and terrestrial systems, it aligns with SDGs 14 and 15, while the use of reproducible and resource-efficient bioinformatic workflows supports SDG 9. In addition, the focus on understudied unicellular lineages promotes ethical and inclusive scientific practice, contributing to a broader and more equitable understanding of biological diversity.

Future work

The results presented in this thesis represent a step towards understanding the evolutionary history of developmental transcription factors, but they also highlight several directions for future research. One immediate extension of this work involves the incorporation of transcriptomic analyses, particularly in unicellular holozoans species, that display aggregative stages and for which we have time-resolved RNA sequencing data available such as *Capsaspora*. Integrating expression profiles across aggregation stages would allow us to infer whether the identified transcription factors are transcriptionally regulated during similar processes to early multicellularity.

In addition, although the current dataset already represents a curated resource, more refinement may be necessary. Additional filtering steps and expanded taxon sampling could improve sequence classification and help resolve remaining ambiguities, particularly within the ANTP class, where phylogenetic resolution remains limited. Addressing these issues will be essential for clarifying the evolutionary relationships of deeply divergent homeobox genes.

Finally, selected SOX sequences identified in this study have been shared with an external laboratory for experimental validation. Ongoing in vitro assays will test whether these sequences have functional properties consistent with SOX transcription factors. The outcome of these experiments will provide an important external validation of the computational predictions and will provide us with further information on the evolutionary interpretation of SOX gene origins in unicellular holozoans.

Chapter 6

Glossary

ANTP class Major class of non-TALE homeobox genes including HOX and NKL subclasses, involved in axial patterning and developmental regulation.

BLAST Basic Local Alignment Search Tool used to identify sequence similarity between biological sequences.

CERS homeobox genes Homeobox genes with class-specific conserved regions involved in developmental regulation.

CUT homeobox genes Homeobox genes containing one or more CUT domains that contribute to transcriptional regulation during development.

Developmental Transcription Factor (DTF) DNA-binding protein involved in regulating gene expression during development, controlling processes such as cell-fate specification, tissue patterning, and differentiation.

EukProt Curated database of predicted proteomes spanning the diversity of eukaryotic life.

HMMER Software suite for detecting protein domains using profile hidden Markov models.

HNF homeobox genes Small class of homeobox genes associated with endoderm-derived tissues and metabolic regulation.

Holozoa Eukaryotic clade comprising Metazoa (animals) and their closest unicellular relatives, including choanoflagellates, filastereans, ichthyosporeans, and corallochytreaans.

Homeobox gene Gene encoding a transcription factor containing a conserved homeodomain of approximately 60 amino acids that mediates sequence-specific DNA binding.

- Homeodomain** Highly conserved DNA-binding domain forming a helix–turn–helix structure characteristic of homeobox transcription factors.
- HOX genes** Subclass of ANTP homeobox genes organized in genomic clusters and responsible for anterior–posterior body-axis patterning in animals.
- IQ-TREE** Software for phylogenetic inference using Maximum Likelihood methods.
- iTOL** Interactive Tree Of Life, a web-based tool for visualization, annotation, and editing of phylogenetic trees.
- LIM homeobox genes** Homeobox transcription factors containing LIM zinc-binding domains that mediate protein–protein interactions and act as regulatory scaffolds.
- ModelFinder** Algorithm implemented in IQ-TREE to select the best-fit substitution model for phylogenetic analysis.
- Multicell Genome Lab (MCG)** Research group at the Institute of Evolutionary Biology specializing in comparative genomics and the origin of multicellularity.
- NKL genes** Subclass of ANTP homeobox genes involved in neural patterning, organogenesis, and cell-type specification, generally not clustered.
- Non-TALE homeobox genes** Superclass of homeobox genes lacking the three-amino-acid insertion characteristic of TALE homeobox genes.
- Pfam** Database of protein families represented by profile hidden Markov models (HMMs).
- Phylogenetic tree** Graphical representation of inferred evolutionary relationships among sequences or taxa.
- POU homeobox genes** Homeobox transcription factors defined by a bipartite POU-specific domain and homeodomain, involved in cell-fate determination and identity maintenance.
- Presence–absence matrix** Binary representation summarizing whether a gene family is present or absent across taxa.
- Proteome** Complete set of predicted protein sequences encoded by a genome.
- SIX / SINE genes** Homeobox genes characterized by the SIX domain, involved in sensory organ development and tissue specification.
- SOX genes** Developmental transcription factors defined by an HMG DNA-binding domain; not homeobox genes but key regulators of cell identity and fate.
- Ultrafast bootstrap (UFBoot)** Statistical method for assessing branch support in phylogenetic trees using resampling; values range from 0 to 100.

Unicellular Holozoa Non-metazoan members of Holozoa used as comparative models to study the evolutionary origin of animal multicellularity.

ZF homeobox genes Homeobox genes containing zinc-finger-associated domains, representing a diverse and evolutionarily ancient regulatory group.

Chapter 7

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Appendix A

Functions and positions of targeted genes

Gene family	Main functions	Position along Holozoa
HOX (ANTP)	Anterior-posterior patterning and specification of segmental identity; regulation of body-axis organization.	Restricted to Metazoa; present in Cnidaria, Placozoa and Bilateria.
NKL (ANTP)	Neural patterning, organogenesis and cell-type specification; often act as transcriptional repressors.	Present across Metazoa, from Porifera and Ctenophora to Bilateria.
LIM	Scaffolding transcription factors mediating protein-protein interactions; neuronal differentiation, tissue patterning and cell identity specification.	Present in Metazoa and unicellular holozoans (Filasterea, Ichthyosporea, Cnolochytreia); absent in Choanoflagellata.
POU	Cell-fate determination and maintenance of cellular identity; combinatorial transcriptional regulation.	Present in Metazoa and unicellular holozoans; originated prior to animal multicellularity.
CUT	Transcriptional regulation during development through combined CUT domains and homeodomain.	Detected in Metazoa; not retained in unicellular holozoans in this study.
HNF	Regulation of endoderm-derived tissues and metabolic processes; specialized transcriptional roles.	Restricted to Metazoa in the analyzed dataset.
SIX / SINE	Sensory organ development and tissue specification; cofactor-mediated regulation.	Present in Metazoa; no unicellular holozoan representatives detected.
ZF	Developmental regulation mediated by distinctive zinc-finger-associated homeobox architectures.	Detected in Metazoa and unicellular holozoans, representing a broader distribution than previously recognized.
CERS	Developmental regulation through class-specific conserved regions combined with a homeodomain.	Present in Metazoa; no unicellular holozoan representatives detected.
SOX	Cell-fate specification and maintenance of cellular identity; interaction with POU transcription factors.	Present in unicellular holozoans and Metazoa; originated prior to animal multicellularity.

Table A.1: Summary of developmental transcription factor families analyzed in this study, including their main functions and their known distribution across holozoan lineages.

Appendix B

Taxon sampling list

Table B.1: List of all the species used for taxon sampling, their taxonomic annotation, and the source of each proteome.

Species	Taxonomy	Source
<i>Saccharomyces cerevisiae</i>	Fungi	https://www.uniprot.org/proteomes/UP000002311
<i>Schizosaccharomyces pombe</i>	Fungi	https://www.uniprot.org/proteomes/UP000002485
<i>Tuber melanosporum</i>	Fungi	https://www.uniprot.org/proteomes/UP000006911
<i>Microbotryum intermedium</i>	Fungi	https://www.uniprot.org/proteomes/UP000198372
<i>Testicularia cyperi</i>	Fungi	https://www.uniprot.org/proteomes/UP000246740
<i>Allomyces macrogynus</i>	Fungi	https://www.uniprot.org/proteomes/UP000054350
<i>Catenaria anguillulae</i>	Fungi	https://www.uniprot.org/proteomes/UP000193411
<i>Powellomyces hirtus</i>	Fungi	https://www.uniprot.org/proteomes/UP000318582
<i>Spizellomyces punctatus</i>	Fungi	https://www.uniprot.org/proteomes/UP000053201
<i>Glomus cerebriforme</i>	Fungi	https://www.uniprot.org/proteomes/UP000265703
<i>Lobosporangium transversale</i>	Fungi	https://www.uniprot.org/proteomes/UP000193648
<i>Basidiobolus meristosporus</i>	Fungi	https://www.uniprot.org/proteomes/UP000193498
<i>Coemansia reversa</i>	Fungi	https://www.uniprot.org/proteomes/UP000242474
<i>Corallochytrium limacisporum</i>	Corallochytreia	MCG
<i>Corallochytrium limacisporum</i>	Corallochytreia	MCG
<i>Corallochytrium limacisporum</i>	Corallochytreia	MCG
<i>Corallochytrium limacisporum</i>	Corallochytreia	MCG
<i>Syssomonas multiformis</i>	Corallochytreia	MCG
<i>Amoebidium parasiticum</i>	Ichthyosporea	MCG
<i>Abeoforma whisleri</i>	Ichthyosporea	MCG
<i>Creolimax fragrantissima</i>	Ichthyosporea	MCG
<i>Chromosfaera perkinsii</i>	Ichthyosporea	MCG

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Table B.1 – continued from previous page

Species	Taxonomy	Source
<i>Ichthyophonus hoferi</i>	Ichthyosporea	MCG
<i>Pirum gemmata</i>	Ichthyosporea	MCG
<i>Sphaeroforma arctica</i>	Ichthyosporea	https://www.uniprot.org/proteomes/UP000054560
<i>Sphaerotechum destruens</i>	Ichthyosporea	MCG
<i>Capsaspora owczarzaki</i>	Filasterea	https://www.uniprot.org/proteomes/UP000008743
<i>Ministeria vibrans</i>	Filasterea	MCG
<i>Pigoraptor chileana</i>	Filasterea	MCG
<i>Pigoraptor vietnamica</i>	Filasterea	MCG
<i>Txikispora philomaios</i>	Filasterea	MCG
<i>Tunicaraptor unikontum</i>	Filasterea	MCG
<i>Savillea parva</i>	Choanoflagellata	MCG; EukProt
<i>Didymoeca costata</i>	Choanoflagellata	MCG; EukProt
<i>Diaphanoeca grandis</i>	Choanoflagellata	MCG; EukProt
<i>Stephanoeca diplocostata</i>	Choanoflagellata	MCG; EukProt
<i>Codosida hollandica</i>	Choanoflagellata	MCG
<i>Microstomoeca roanoka</i>	Choanoflagellata	EukProt
<i>Salpingoeca rosetta</i>	Choanoflagellata	https://www.uniprot.org/proteomes/UP000007799 ; EukProt
<i>Choanoeca perplexa</i>	Choanoflagellata	MCG; EukProt
<i>Monosiga brevicollis</i>	Choanoflagellata	https://www.uniprot.org/proteomes/UP000001357 ; EukProt
<i>Mylnosiga fluctuans</i>	Choanoflagellata	MCG; EukProt
<i>Salpingoeca helianthica</i>	Choanoflagellata	MCG; EukProt
<i>Salpingoeca infusorum</i>	Choanoflagellata	MCG
<i>Salpingoeca kjevrii</i>	Choanoflagellata	EukProt
<i>Salpingoeca urceolata</i>	Choanoflagellata	MCG; EukProt
<i>Salpingoeca macrocollata</i>	Choanoflagellata	MCG; EukProt
<i>Salpingoeca punica</i>	Choanoflagellata	MCG; EukProt
<i>Salpingoeca dolichotechata</i>	Choanoflagellata	MCG
<i>Hartaetosiga balthica</i>	Choanoflagellata	MCG; EukProt
<i>Hartaetosiga gracilis</i>	Choanoflagellata	MCG; EukProt
<i>Bicosta minor</i>	Choanoflagellata	EukProt
<i>Choanoeca flexa</i>	Choanoflagellata	MCG; EukProt
<i>Barroeca monosierra</i>	Choanoflagellata	EukProt
<i>Salpingoeca helianthica</i>	Choanoflagellata	MCG; EukProt
<i>Acanthoea spectabilis</i>	Choanoflagellata	MCG; EukProt

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Table B.1 – continued from previous page

Species	Taxonomy	Source
<i>Acanthoea spectabilis</i>	Choanoflagellata	MCG; EukProt
<i>Choanoeca flexa</i>	Choanoflagellata	MCG; EukProt
<i>Choanoeca perplexa</i>	Choanoflagellata	MCG; EukProt
<i>Codosida hollandica</i>	Choanoflagellata	MCG
<i>Diaphanoeca grandis</i>	Choanoflagellata	MCG; EukProt
<i>Didymoeca costata</i>	Choanoflagellata	MCG; EukProt
<i>Hartaetosiga gracilis</i>	Choanoflagellata	MCG; EukProt
<i>Hartaetosiga balthica</i>	Choanoflagellata	MCG; EukProt
<i>Helgoeca nana</i>	Choanoflagellata	MCG; EukProt
<i>Mylnosiga fluctuans</i>	Choanoflagellata	MCG; EukProt
<i>Monosiga brevicollis</i>	Choanoflagellata	https://www.uniprot.org/proteomes/UP000001357 ; EukProt
<i>Salpingoeca rosetta</i>	Choanoflagellata	https://www.uniprot.org/proteomes/UP000007799 ; EukProt
<i>Salpingoeca dolichotechata</i>	Choanoflagellata	MCG
<i>Salpingoeca infusorum</i>	Choanoflagellata	MCG
<i>Salpingoeca macrocollata</i>	Choanoflagellata	MCG; EukProt
<i>Salpingoeca punica</i>	Choanoflagellata	MCG; EukProt
<i>Salpingoeca urceolata</i>	Choanoflagellata	MCG; EukProt
<i>Savillea parva</i>	Choanoflagellata	MCG; EukProt
<i>Stephanoeca diplocostata</i>	Choanoflagellata	MCG; EukProt
<i>Ephydatia muelleri</i>	Porifera	https://figshare.com/articles/dataset/Mirror_Ephydatia_muelleri_Supplementary_Files_Kenny_et_al_2020_/11847195?file=21711594
<i>Chondrosia reniformis</i>	Porifera	https://datadryad.org/stash/dataset/doi:10.5061/dryad.dncjsxm47
<i>Tethya wilhelma</i>	Porifera	https://bitbucket.org/molpalmuc/tethya_wilhelma-genome/src/master/gene_sets/twilhelma_v01_augustus_prots.fasta.gz
<i>Petrosia ficiformis</i>	Porifera	https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/24737
<i>Petrosia ficiformis</i>	Porifera	https://datadryad.org/stash/dataset/doi:10.5061/dryad.dncjsxm47
<i>Aphrocallistes vastus</i>	Porifera	https://era.library.ualberta.ca/items/3df7eb24-1f79-43d9-bcb4-49ef043ea226#.WEbM_tFVKlM
<i>Oopsacas minuta</i>	Porifera	https://www.ncbi.nlm.nih.gov/datasets/taxonomy/111878/
<i>Leucosolenia complicata</i>	Porifera	https://compagen.unit.oist.jp/datasets.html
<i>Sycon ciliatum</i>	Porifera	https://compagen.unit.oist.jp/datasets.html

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Table B.1 – continued from previous page

Species	Taxonomy	Source
Amphimedon queenslandica	Porifera	https://www.uniprot.org/proteomes/UP000007879
Spongilla lacustris	Porifera	https://zenodo.org/record/7789803
Hormiphora californensis	Ctenophora	https://github.com/conchoecia/hormiphora/tree/master/annotation/proteins/prottrans_predict
Pleurobrachia bachei	Ctenophora	EukProt
Pleurobrachia bachei	Ctenophora	https://neurobase.rc.ufl.edu/pleurobrachia/download
Beroe abyssicola	Ctenophora	https://neurobase.rc.ufl.edu/pleurobrachia/download
Bolinopsis microptera	Ctenophora	https://datadryad.org/stash/dataset/doi:10.5061/dryad.dncjsxm47
Pukia falcata	Ctenophora	https://neurobase.rc.ufl.edu/pleurobrachia/download
Mnemiopsis leidyi	Ctenophora	https://research.nhgri.nih.gov/mnemiopsis/download/download.cgi?dl=proteome
Hoilungia hongkongensis	Placozoa	https://bitbucket.org/molpalmuc/hoilungia-genome/src/master/sequences/Hhon_BRAKER1_proteins.fasta.gz
Trichoplax sp.	Placozoa	https://www.uniprot.org/proteomes/UP000253843
Trichoplax adhaerens	Placozoa	https://www.uniprot.org/proteomes/UP000009022
Aurelia aurita	Cnidaria	https://marinegenomics.oist.jp/aurelia_aurita/viewer/download?project_id=69
Rhopilema esculentum	Cnidaria	https://datadryad.org/stash/dataset/doi:10.5061/dryad.dncjsxm47
Hydra magnipapillata	Cnidaria	https://research.nhgri.nih.gov/hydra/download/?dl=aa
Clytia hemisphaerica	Cnidaria	https://www.uniprot.org/proteomes/UP000594262
Nematostella vectensis	Cnidaria	https://www.uniprot.org/proteomes/UP000001593
Pocillopora damicornis	Cnidaria	https://www.uniprot.org/proteomes/UP000275408
Exaiptasia diaphana	Cnidaria	https://www.uniprot.org/proteomes/UP000887567
Stylophora pistillata	Cnidaria	https://www.uniprot.org/proteomes/UP000225706
Hydractinia echinata	Cnidaria	https://research.nhgri.nih.gov/hydractinia/download/index.cgi?dl=pm
Morbakka virulenta	Cnidaria	https://marinegenomics.oist.jp/morbakka_virulenta/viewer/download?project_id=70
Actinia tenebrosa	Cnidaria	https://www.uniprot.org/proteomes/UP000515163
Apis mellifera	Bilateria	https://www.uniprot.org/proteomes/UP000005203
Drosophila melanogaster	Bilateria	https://www.uniprot.org/proteomes/UP000000803
Lingula unguis	Bilateria	https://www.uniprot.org/proteomes/UP000085678
Danio rerio	Bilateria	https://www.uniprot.org/proteomes/UP000000437
Homo sapiens	Bilateria	https://www.uniprot.org/proteomes/UP000005640

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Species	Taxonomy	Source
Crassostrea gigas	Bilateria	https://ftp.ncbi.nlm.nih.gov/genomes/refseq/invertebrate/Crassostrea_gigas/latest_assembly_versions/GCF_902806645.1_cgigas_uk_roslin_v1/
Caenorhabditis elegans	Bilateria	https://www.uniprot.org/proteomes/UP000001940

Appendix C

List of sequences used as query for each Gene Family

- CERS**
- CERS2 (ceramide synthase 2) (*Homo sapiens*).
 - CERS3 (ceramide synthase 3 isoform 2) (*Homo sapiens*).
 - CERS4 (ceramide synthase 4) (*Homo sapiens*).
 - CERS5 (ceramide synthase 5) (*Homo sapiens*).
 - CERS6 (ceramide synthase 6) (*Homo sapiens*).
- CUT**
- SATB2 (DNA-binding protein SATB2) (*Homo sapiens*).
 - CUX1 (homeobox protein cut-like 1 isoform 1) (*Homo sapiens*).
 - CUX2 (homeobox protein cut-like 2 isoform 2) (*Homo sapiens*).
 - ONECUT1 (one cut domain family member 1) (*Homo sapiens*).
 - ONECUT2 (one cut domain family member 2 isoform 1) (*Homo sapiens*).
 - SATB1 (DNA-binding protein SATB1 isoform 1) (*Homo sapiens*).
 - ONECUT3 (one cut domain family member 3) (*Homo sapiens*).
- HNF**
- HMBOX1 (homeobox-containing protein 1 isoform 1) (*Homo sapiens*).
 - HNF1A (hepatocyte nuclear factor 1-alpha isoform 1) (*Homo sapiens*).
 - HNF1B (hepatocyte nuclear factor 1-beta isoform 1) (*Homo sapiens*).
- HOX**
- EVX1 (homeobox even-skipped homolog protein 1 isoform 1) (*Homo sapiens*).
 - EVX2 (even-skipped homeobox 2) (*Homo sapiens*).
 - HOXA1 (homeobox protein Hox-A1 isoform 1) (*Homo sapiens*).
 - HOXA11 (homeobox protein Hox-A11 isoform 1) (*Homo sapiens*).
 - HOXB1 (homeobox protein Hox-B1 isoform 1) (*Homo sapiens*).
 - HOXB9 (homeobox protein Hox-B9 isoform 1) (*Homo sapiens*).
 - HOXC10 (homeobox protein Hox-C10 isoform 1) (*Homo sapiens*).
 - HOXC13 (homeobox protein Hox-C13 isoform 1) (*Homo sapiens*).
 - HOXD11 (homeobox protein Hox-D11 isoform 1) (*Homo sapiens*).
 - HOXD13 (homeobox protein Hox-D13 isoform 1) (*Homo sapiens*).
 - MEIS1 (homeobox protein Meis1 isoform 1) (*Homo sapiens*).
 - PKNOX1 (PBX/knotted 1 homeobox protein 1 isoform 1) (*Homo sapiens*).
 - PBX1 (pre-B-cell leukemia transcription factor 1 isoform 1) (*Homo sapiens*).

- LIM**
- LMX1A (LIM homeobox transcription factor 1 alpha) (*Homo sapiens*).
 - LMX1B (LIM homeobox transcription factor 1 beta isoform 1) (*Homo sapiens*).
 - LHX6 (LIM homeobox protein Lhx6 isoform 1) (*Homo sapiens*).
 - LHX1 (LIM homeobox protein 1) (*Homo sapiens*).
 - LHX2 (LIM homeobox protein 2 isoform 1) (*Homo sapiens*).
 - LHX3 (LIM/homeobox protein Lhx3 isoform 1) (*Homo sapiens*).
 - LHX4 (LIM/homeobox protein Lhx4 isoform 1) (*Homo sapiens*).
 - LHX5 (LIM/homeobox protein Lhx5 isoform 1) (*Homo sapiens*).
 - LHX8 (LIM homeobox protein 8 isoform 1) (*Homo sapiens*).
 - LHX9 (LIM homeobox protein 9 isoform 1) (*Homo sapiens*).
 - ISL1 (ISL LIM homeobox 1 isoform 1) (*Homo sapiens*).
 - ISL2 (ISL LIM homeobox 2 isoform 1) (*Homo sapiens*).
- NKL**
- BARH1 (barH-like 1 homeobox protein) (*Homo sapiens*).
 - BARHL2 (barH-like 2 homeobox protein isoform 1) (*Homo sapiens*).
 - DLX1 (distal-less homeobox 1 isoform 1) (*Homo sapiens*).
 - DLX2 (homeobox protein DLX-2 isoform 1) (*Homo sapiens*).
 - DLX3 (homeobox protein DLX-3 isoform 1) (*Homo sapiens*).
 - DLX4 (homeobox protein DLX-4 isoform 1) (*Homo sapiens*).
 - DLX5 (homeobox protein DLX-5 isoform 1) (*Homo sapiens*).
 - DLX6 (homeobox protein DLX-6 isoform 1) (*Homo sapiens*).
 - EMX1 (empty spiracles homeobox 1 isoform 1) (*Homo sapiens*).
 - EMX2 (empty spiracles homeobox 2 isoform 1) (*Homo sapiens*).
 - EN1 (engrailed homeobox 1 isoform 1) (*Homo sapiens*).
 - EN2 (engrailed homeobox 2 isoform 1) (*Homo sapiens*).
 - GSX1 (genetic screen homeobox 1 isoform 1) (*Homo sapiens*).
 - GSX2 (genetic screen homeobox 2 isoform 1) (*Homo sapiens*).
 - HHX (hematopoietically expressed homeobox isoform 1) (*Homo sapiens*).
 - HLX (H2.0-like homeobox protein isoform 1) (*Homo sapiens*).
 - MSX1 (homeobox protein MSX-1 isoform 1) (*Homo sapiens*).
 - MSX2 (homeobox protein MSX-2 isoform 1) (*Homo sapiens*).
 - NKX2-1 (homeobox protein Nkx-2.1 isoform 1) (*Homo sapiens*).
 - NKX2-5 (homeobox protein Nkx-2.5 isoform 1) (*Homo sapiens*).
 - NKX3-1 (homeobox protein Nkx-3.1 isoform 1) (*Homo sapiens*).
 - TLX1 (T-cell leukemia homeobox protein 1 isoform 1) (*Homo sapiens*).
- POU**
- POU domain class 4 transcription factor 1 (*Homo sapiens*).
 - pituitary-specific positive transcription factor 1 isoform alpha (*Homo sapiens*).
 - POU domain class 1 transcription factor 1 isoform 1 (*Homo sapiens*).
 - POU domain class 1 transcription factor 2 isoform 1 (*Homo sapiens*).
 - POU domain class 1 transcription factor 3 isoform 1 (*Homo sapiens*).
 - POU domain class 2 transcription factor 1 isoform 1 (*Homo sapiens*).
 - POU domain class 2 transcription factor 2 isoform 1 (*Homo sapiens*).
 - POU domain class 2 transcription factor 3 isoform 1 (*Homo sapiens*).
 - POU domain class 3 transcription factor 1 isoform 1 (*Homo sapiens*).
 - POU domain class 3 transcription factor 2 isoform 1 (*Homo sapiens*).
 - POU domain class 3 transcription factor 3 isoform 1 (*Homo sapiens*).

- POU domain class 3 transcription factor 4 isoform 1 (*Homo sapiens*).
 - POU domain class 4 transcription factor 2 isoform 1 (*Homo sapiens*).
 - POU domain class 4 transcription factor 3 isoform 1 (*Homo sapiens*).
 - POU domain class 5 transcription factor 1 isoform 1 (*Homo sapiens*).
- SINE**
- SIX1 (homeobox protein SIX1 isoform 1) (*Homo sapiens*).
 - SIX2 (homeobox protein SIX2 isoform 1) (*Homo sapiens*).
 - SIX3 (homeobox protein SIX3 isoform 1) (*Homo sapiens*).
 - SIX4 (homeobox protein SIX4 isoform 1) (*Homo sapiens*).
 - SIX5 (homeobox protein SIX5 isoform 1) (*Homo sapiens*).
 - SIX6 (homeobox protein SIX6 isoform 1) (*Homo sapiens*).
- SOX**
- SOX1 (SRY-box transcription factor 1 isoform 1) (*Homo sapiens*).
 - SOX2 (SRY-box transcription factor 2 isoform 1) (*Homo sapiens*).
 - SOX3 (SRY-box transcription factor 3 isoform 1) (*Homo sapiens*).
 - SOX4 (SRY-box transcription factor 4 isoform 1) (*Homo sapiens*).
 - SOX5 (SRY-box transcription factor 5 isoform 1) (*Homo sapiens*).
 - SOX6 (SRY-box transcription factor 6 isoform 1) (*Homo sapiens*).
 - SOX7 (transcription factor SOX-7) (*Homo sapiens*).
 - SOX8 (SRY-box transcription factor 8 isoform 1) (*Homo sapiens*).
 - SOX9 (SRY-box transcription factor 9 isoform 1) (*Homo sapiens*).
 - SOX10 (SRY-box transcription factor 10 isoform 1) (*Homo sapiens*).
 - SOX11 (SRY-box transcription factor 11 isoform 1) (*Homo sapiens*).
 - SOX12 (SRY-box transcription factor 12 isoform 1) (*Homo sapiens*).
 - SOX13 (SRY-box transcription factor 13 isoform 1) (*Homo sapiens*).
 - SOX14 (SRY-box transcription factor 14 isoform 1) (*Homo sapiens*).
 - SOX15 (SRY-box transcription factor 15 isoform 1) (*Homo sapiens*).
 - SOX17 (SRY-box transcription factor 17 isoform 1) (*Homo sapiens*).
 - SOX18 (SRY-box transcription factor 18 isoform 1) (*Homo sapiens*).
 - SOX21 (SRY-box transcription factor 21 isoform 1) (*Homo sapiens*).
 - SOX30 (SRY-box transcription factor 30 isoform 1) (*Homo sapiens*).
 - SRY (sex-determining region Y protein) (*Homo sapiens*).
- ZF**
- HOMEZ (homeobox and leucine zipper protein Homez) (*Homo sapiens*).
 - ZEB1 (zinc finger E-box-binding homeobox 1 isoform 1) (*Homo sapiens*).
 - ZEB2 (zinc finger E-box-binding homeobox 2 isoform 1) (*Homo sapiens*).
 - ZFHX2 (zinc finger homeobox protein 2 isoform 1) (*Homo sapiens*).
 - ZFHX3 (zinc finger homeobox protein 3 isoform 1) (*Homo sapiens*).
 - ZFHX4 (zinc finger homeobox protein 4 isoform 1) (*Homo sapiens*).
 - ZHX1 (zinc fingers and homeoboxes protein 1 isoform 1) (*Homo sapiens*).
 - ZHX2 (zinc fingers and homeoboxes protein 2 isoform 1) (*Homo sapiens*).
 - ZHX3 (zinc fingers and homeoboxes protein 3 isoform 1) (*Homo sapiens*).
 - ALX1 (aristaless-like homeobox 1 isoform 1) (*Homo sapiens*).
 - ALX3 (aristaless-like homeobox 3 isoform 1) (*Homo sapiens*).
 - ALX4 (aristaless-like homeobox 4 isoform 1) (*Homo sapiens*).
 - TSHZ1 (teashirt zinc finger homeobox 1 isoform 1) (*Homo sapiens*).

Appendix D

R script used to analyze and filter phylogenetic trees and create the final dataset

```
# Libraries
library(ape)
library(dplyr)
library(ggplot2)
library(reshape2)
library(pheatmap)
library(tidyr)
library(tibble)
library(stringr)
library(scales)
library(ggthemes)
library(paletteer)

# Phylo trees
LIM <- "LIM_final_doms_phylogeny_treefile"
LIM_tree <- read.tree(LIM)
CERS <- "CERS_final_doms_phylogeny_treefile"
CERS_tree <- read.tree(CERS)
CUT <- "CUT_final_doms_phylogeny_treefile"
CUT_tree <- read.tree(CUT)
HNF <- "HNF_final_doms_phylogeny_treefile"
HNF_tree <- read.tree(HNF)
POU <- "POU_final_doms_phylogeny_treefile"
POU_tree <- read.tree(POU)
SINE <- "SINE_final_doms_phylogeny_treefile"
SINE_tree <- read.tree(SINE)
SOX <- "SOX_final_doms_phylogeny_treefile"
SOX_tree <- read.tree(SOX)
HOX <- "HOX_final_doms_phylogeny_treefile"
HOX_tree <- read.tree(HOX)
NKL <- "NKL_final_doms_phylogeny_treefile"
NKL_tree <- read.tree(NKL)
ZF <- "ZF_final_doms_phylogeny_treefile"
ZF_tree <- read.tree(ZF)

# Species reference code csv
reference_code <- read.csv("code.csv", header = TRUE, sep = ";")
reference_code[4:30] <- list(NULL)
reference_code <- reference_code[-c(163:1182), ]

reference_code <- rows.insert(reference_code, tibble(Species = "Microstomoeca roanoka",
```



```

Taxonomy = "Choanoflagellata", Code = "Mroan"))
reference_code <- rows.insert(reference_code, tibble(Code = "Skvev", Species = "Salpingoeca
kvevrii", Taxonomy = "Choanoflagellata"))
reference_code <- rows.insert(reference_code, tibble(Code = "Bmino", Species = "Bicosta minor
", Taxonomy = "Choanoflagellata"))
reference_code <- rows.insert(reference_code, tibble(Code = "Bmono", Species = "Barroeca
monosierra", Taxonomy = "Choanoflagellata"))

# LIM
LIM_leaf <- LIM_tree$tip.label
LIM_dataset <- data.frame(Code=substr(LIM_leaf, 1, 5), Gene="LIM", Header=LIM_leaf)
LIM_dataset <- LIM_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
LIM_dataset <- LIM_dataset[-c(as.numeric(rownames(LIM_dataset[apply(LIM_dataset, 1, function(x
) any(is.na(x))), ]))), ]
# CERS
CERS_leaf <- CERS_tree$tip.label
CERS_dataset <- data.frame(Code=substr(CERS_leaf, 1, 5), Gene="CERS", Header=CERS_leaf)
CERS_dataset <- CERS_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
CERS_dataset <- CERS_dataset[-c(as.numeric(rownames(CERS_dataset[apply(CERS_dataset, 1,
function(x) any(is.na(x))), ]))), ]
# CUT
CUT_leaf <- CUT_tree$tip.label
CUT_dataset <- data.frame(Code=substr(CUT_leaf, 1, 5), Gene="CUT", Header=CUT_leaf)
CUT_dataset <- CUT_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
CUT_dataset <- CUT_dataset[-c(as.numeric(rownames(CUT_dataset[apply(CUT_dataset, 1, function(x
) any(is.na(x))), ]))), ]
# HNF
HNF_leaf <- HNF_tree$tip.label
HNF_dataset <- data.frame(Code=substr(HNF_leaf, 1, 5), Gene="HNF", Header=HNF_leaf)
HNF_dataset <- HNF_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
HNF_dataset <- HNF_dataset[-c(as.numeric(rownames(HNF_dataset[apply(HNF_dataset, 1, function(x
) any(is.na(x))), ]))), ]
# POU
POU_leaf <- POU_tree$tip.label
POU_dataset <- data.frame(Code=substr(POU_leaf, 1, 5), Gene="POU", Header=POU_leaf)
POU_dataset <- POU_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
POU_dataset <- POU_dataset[-c(as.numeric(rownames(POU_dataset[apply(POU_dataset, 1, function(x
) any(is.na(x))), ]))), ]
# SINE
SINE_leaf <- SINE_tree$tip.label
SINE_dataset <- data.frame(Code=substr(SINE_leaf, 1, 5), Gene="SINE", Header=SINE_leaf)
SINE_dataset <- SINE_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
SINE_dataset <- SINE_dataset[-c(as.numeric(rownames(SINE_dataset[apply(SINE_dataset, 1,
function(x) any(is.na(x))), ]))), ]

# Indicate the number of closer nodes it can search
K <- 4

# SOX
queries_SOX <- c("sox")
SOX_leaf_queries <- grep(paste(queries_SOX, collapse = "|"),
  SOX_tree$tip.label,
  value = TRUE,
  ignore.case = TRUE)

unit_tree <- SOX_tree
unit_tree$edge.length <- rep(1, nrow(unit_tree$edge))
D <- dist.nodes(unit_tree)
ntip <- length(unit_tree$tip.label)
dist_to_mrca_edges <- function(tree, tip1, tip2, Dmat) {
  m <- getMRCA(tree, c(tip1, tip2))
  if (is.na(m)) return(list(d1=NA_real_, d2=NA_real_))
  i1 <- match(tip1, tree$tip.label)

```

```

    i2 <- match(tip2, tree$tip.label)
    list(d1 = Dmat[i1, m], d2 = Dmat[i2, m])
}
all_tips <- unit_tree$tip.label
pairs <- expand.grid(query = SOX_leaf_queries,
                    tip = all_tips,
                    stringsAsFactors = FALSE)
withinK <- logical(nrow(pairs))
for (i in seq_len(nrow(pairs))) {
  q <- pairs$query[i]
  t <- pairs$tip[i]
  dd <- dist.to_mrca_edges(unit_tree, q, t, D)
  withinK[i] <- !any(is.na(unlist(dd))) && max(dd$d1, dd$d2) <= K
}
SOX_tips_withinK <- unique(pairs$tip[withinK])
SOX_dataset <- data.frame(Code=substr(SOX_tips_withinK, 1, 5), Gene="SOX", Header=
  SOX_tips_withinK)
SOX_dataset <- SOX_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
SOX_dataset <- SOX_dataset[-c(as.numeric(rownames(SOX_dataset[apply(SOX_dataset, 1, function(x)
  any(is.na(x))), ]))), ]

# HOX
queries_HOX <- c("cdx", "evx", "gbx", "gsx", "hoxa", "mnx", "meox", "pdx", "hox", "labial", "
  lab", "abdominal", "abd", "bicoid", "bcd")
HOX_leaf_queries <- grep(paste(queries_HOX, collapse = "|"),
  HOX_tree$tip.label,
  value = TRUE,
  ignore.case = TRUE)

unit_tree <- HOX_tree
unit_tree$edge.length <- rep(1, nrow(unit_tree$edge))
D <- dist.nodes(unit_tree)
ntip <- length(unit_tree$tip.label)
dist.to_mrca_edges <- function(tree, tip1, tip2, Dmat) {
  m <- getMRCA(tree, c(tip1, tip2))
  if (is.na(m)) return(list(d1=NA_real_, d2=NA_real_))
  i1 <- match(tip1, tree$tip.label)
  i2 <- match(tip2, tree$tip.label)
  list(d1 = Dmat[i1, m], d2 = Dmat[i2, m])
}
all_tips <- unit_tree$tip.label
pairs <- expand.grid(query = HOX_leaf_queries,
                    tip = all_tips,
                    stringsAsFactors = FALSE)
withinK <- logical(nrow(pairs))
for (i in seq_len(nrow(pairs))) {
  q <- pairs$query[i]
  t <- pairs$tip[i]
  dd <- dist.to_mrca_edges(unit_tree, q, t, D)
  withinK[i] <- !any(is.na(unlist(dd))) && max(dd$d1, dd$d2) <= K
}
HOX_tips_withinK <- unique(pairs$tip[withinK])
HOX_dataset <- data.frame(Code=substr(HOX_tips_withinK, 1, 5), Gene="HOX", Header=
  HOX_tips_withinK)
HOX_dataset <- HOX_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
HOX_dataset <- HOX_dataset[-c(as.numeric(rownames(HOX_dataset[apply(HOX_dataset, 1, function(x)
  any(is.na(x))), ]))), ]

# NKL
queries_NKL <- c("BARHL", "BARX", "BSX", "DBX", "DLX", "EMX", "HHEX", "HLX", "LBX", "NANOG", "
  NKX", "NOTO", "TLX", "VAX", "VENT", "NK", "distal", "empty_spiracles", "scarecrow", "scro
  ")
NKL_leaf_queries <- grep(paste(queries_NKL, collapse = "|"),
  NKL_tree$tip.label,
  value = TRUE,
  ignore.case = TRUE)

unit_tree <- NKL_tree

```

```

unit_tree$edge.length <- rep(1, nrow(unit_tree$edge))
D <- dist.nodes(unit_tree)
ntip <- length(unit_tree$tip.label)
dist_to_mrca_edges <- function(tree, tip1, tip2, Dmat) {
  m <- getMRCA(tree, c(tip1, tip2))
  if (is.na(m)) return(list(d1=NA_real_, d2=NA_real_))
  i1 <- match(tip1, tree$tip.label)
  i2 <- match(tip2, tree$tip.label)
  list(d1 = Dmat[i1, m], d2 = Dmat[i2, m])
}
all_tips <- unit_tree$tip.label
pairs <- expand.grid(query = NKL_leaf_queries,
                    tip = all_tips,
                    stringsAsFactors = FALSE)
withinK <- logical(nrow(pairs))
for (i in seq_len(nrow(pairs))) {
  q <- pairs$query[i]
  t <- pairs$tip[i]
  dd <- dist_to_mrca_edges(unit_tree, q, t, D)
  withinK[i] <- !any(is.na(unlist(dd))) && max(dd$d1, dd$d2) <= K
}
NKL_tips_withinK <- unique(pairs$tip[withinK])
NKL_dataset <- data.frame(Code=substr(NKL_tips_withinK, 1, 5), Gene="NKL", Header=
  NKL_tips_withinK)
NKL_dataset <- NKL_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
NKL_dataset <- NKL_dataset[-c(as.numeric(rownames(NKL_dataset[apply(NKL_dataset, 1, function(x)
  any(is.na(x))), ]))), ]

# ZF
queries_ZF <- c("adnp", "tshz", "zeb", "zfhx", "homez", "teashirt", "zinc_finger")
ZF_leaf_queries <- grep(paste(queries_ZF, collapse = "|"),
  ZF_tree$tip.label,
  value = TRUE,
  ignore.case = TRUE)

unit_tree <- ZF_tree
unit_tree$edge.length <- rep(1, nrow(unit_tree$edge))
D <- dist.nodes(unit_tree)
ntip <- length(unit_tree$tip.label)
dist_to_mrca_edges <- function(tree, tip1, tip2, Dmat) {
  m <- getMRCA(tree, c(tip1, tip2))
  if (is.na(m)) return(list(d1=NA_real_, d2=NA_real_))
  i1 <- match(tip1, tree$tip.label)
  i2 <- match(tip2, tree$tip.label)
  list(d1 = Dmat[i1, m], d2 = Dmat[i2, m])
}
all_tips <- unit_tree$tip.label
pairs <- expand.grid(query = ZF_leaf_queries,
                    tip = all_tips,
                    stringsAsFactors = FALSE)
withinK <- logical(nrow(pairs))
for (i in seq_len(nrow(pairs))) {
  q <- pairs$query[i]
  t <- pairs$tip[i]
  dd <- dist_to_mrca_edges(unit_tree, q, t, D)
  withinK[i] <- !any(is.na(unlist(dd))) && max(dd$d1, dd$d2) <= K
}
ZF_tips_withinK <- unique(pairs$tip[withinK])
ZF_dataset <- data.frame(Code=substr(ZF_tips_withinK, 1, 5), Gene="ZF", Header=ZF_tips_withinK
  )
ZF_dataset <- ZF_dataset %>%
  full_join(reference_code, by = c("Code" = "Code"));
ZF_dataset <- ZF_dataset[-c(as.numeric(rownames(ZF_dataset[apply(ZF_dataset, 1, function(x)
  any(is.na(x))), ]))), ]

# Build full dataset
dataset <- rbind(LIM_dataset, CERS_dataset, POU_dataset, SOX_dataset, NKL_dataset, HOX_dataset
  , ZF_dataset, SINE_dataset, HNF_dataset, CUT_dataset)

```

```
# Taxonomy subsets
Non_Bilateria <- c("Porifera", "Ctenophora", "Cnidaria", "Placozoa")
Bilateria <- c("Arthropoda", "Nematoda", "Priapulida", "Tardigrada", "Mollusca", "Annelida", "Platyhelminthes", "Rotifera", "Nemertea", "Bryozoa", "Phoronida", "Brachiopoda", "Chordata", "Chordata", "Echinodermata", "Hemichordata", "Acoelomorpha", "Xenoturbellida")
Unicellular_Holozoa <- c("Choanoflagellata", "Ichthyosporea", "Filasterea", "Corallochytreia")
Fungi <- c("Ascomycota", "Basidiomycota", "Mucoromycota", "Zoopagomycota", "Chytridiomycota", "Blastocladiomycota")
dataset <- dataset %>%
  mutate(Classification = case_when(
    Taxonomy %in% Non_Bilateria ~ "Non_Bilateria",
    Taxonomy %in% Bilateria ~ "Bilateria",
    Taxonomy %in% Unicellular_Holozoa ~ "Unicellular_Holozoa",
    Taxonomy %in% Fungi ~ "Fungi",
    TRUE ~ NA_character_
  ))
dataset <- dataset %>%
  mutate(
    Classification = factor(
      Classification,
      levels = c("Bilateria", "Non_Bilateria", "Unicellular_Holozoa", "Fungi")
    )
  ) %>%
  arrange(Classification, Taxonomy) %>% # orders rows by Classification first
  mutate(Taxonomy = factor(Taxonomy, levels = unique(Taxonomy)))

# Duplicated headers
duplications <- dataset[duplicated(dataset$Header) | duplicated(dataset$Header, fromLast = TRUE), ]

# Cleaning LIM
duplications_LIM_clean <- duplications[
  duplications$Header %in% duplications$Header[duplications$Gene == "LIM"] &
  duplications$Gene != "LIM",
]
# Cleaning CUT
duplications_CUT_clean <- duplications[
  duplications$Header %in% duplications$Header[duplications$Gene == "CUT"] &
  duplications$Gene != "CUT",
]
# Cleaning POU
duplications_POU_clean <- duplications[
  duplications$Header %in% duplications$Header[duplications$Gene == "POU"] &
  duplications$Gene != "POU",
]
# Cleaning SINE
duplications_SINE_clean <- duplications[
  duplications$Header %in% duplications$Header[duplications$Gene == "SINE"] &
  duplications$Gene != "SINE",
]
# Cleaning HOX
duplications_HOX <- duplications[
  grepl(paste(queries_HOX, collapse = "|"),
    duplications$Header,
    ignore.case = TRUE),
]
duplications_HOX_clean <- duplications_HOX[duplications_HOX$Gene != "HOX", ]
# Cleaning NKL
duplications_NKL <- duplications[
  grepl(paste(queries_NKL, collapse = "|"),
    duplications$Header,
    ignore.case = TRUE),
]
duplications_NKL_clean <- duplications_NKL[duplications_NKL$Gene != "NKL", ]
# Cleaning ZF
duplications_ZF <- duplications[
  grepl(paste(queries_ZF, collapse = "|"),
```

```

        duplications$Header ,
        ignore.case = TRUE),
]
duplications_ZF_clean <- duplications_ZF[duplications_ZF$Gene!="ZF", ]

# Joining duplications
duplications_to_clean <- rbind(duplications_LIM_clean, duplications_CUT_clean,
    duplications_POU_clean, duplications_SINE_clean, duplications_HOX_clean,
    duplications_ZF_clean, duplications_NKL_clean)

# Delete duplications
dataset_clean <- anti_join(dataset, duplications_to_clean,
    by = colnames(dataset))
duplications_new <- dataset_clean[duplicated(dataset_clean$Header) | duplicated(
    dataset_clean$Header, fromLast = TRUE), ]
dataset_clean <- anti_join(dataset_clean, duplications_new,
    by = colnames(dataset_clean))

```

Appendix E

Unedited phylogenetic trees

Gene	Inference method	Software	Substitution model	Branch support
LIM	Maximum Likelihood	IQ-TREE	Q.insect+F+I+G4	UFBoot (1000)
CERS	Maximum Likelihood	IQ-TREE	Q.Pfam+I+G4	UFBoot (1000)
POU	Maximum Likelihood	IQ-TREE	Q.insect+F+I+G4	UFBoot (1000)
SINE	Maximum Likelihood	IQ-TREE	Q.insect+F+I+G4	UFBoot (1000)
CUT	Maximum Likelihood	IQ-TREE	Q.insect+F+G4	UFBoot (1000)
HNF	Maximum Likelihood	IQ-TREE	Q.insect+F+I+G4	UFBoot (1000)
HOX	Maximum Likelihood	IQ-TREE	Q.insect+F+G4	UFBoot (1000)
NKL	Maximum Likelihood	IQ-TREE	Q.insect+F+G4	UFBoot (1000)
ZF	Maximum Likelihood	IQ-TREE	Q.insect+F+G4	UFBoot (1000)
SOX	Maximum Likelihood	IQ-TREE	Q.insect+F+G4	UFBoot (1000)

Table E.1: Summary of phylogenetic analyses performed for each targeted gene family. All trees were inferred using a Maximum Likelihood approach in IQ-TREE, with model selection performed using ModelFinder and branch support assessed via ultrafast bootstrap (UFBoot).

The files for the trees can be found at the Github repository:

https://github.com/dimontia/TFM_Diego_Montiel_Ayora/tree/main/Dataset_Script

Appendix F

Github repository

https://github.com/dimontia/TFM_Diego_Montiel_Ayora